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VHDVSDefTopic

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About the UltraVS256

About the UltraVS256

The UltraVS256 (UVS256) Voltage Supply is a high-density power supply with a high channel count and merging capability that allows you to test devices with current consumption up to 800mA that can be merged up to 5.6A. This instrument board provides 256 device power supplies per UltraFLEX slot with an output voltage range of from -2V up to 18V at up to 800mA.

The UVS256 High Accuracy includes all features of the UVS256 and improves force voltage accuracy. In this description, the term “UVS256” refers to the UVS256 and the UVS256 High Accuracy unless otherwise noted.

Each UVS256 instrument includes two rider channel boards, each with 128 channels. Each channel board allows up to 64 HC (high-current) channels, and any channels on a board can be HC channels. This means that the UVS256 can provide up to 128 HC (high-current) supplies, operating at up to 800mA with voltage limits of -2V to 2.5V.

Channels not used as HC are standard current (SC). This means that each UVS256 instrument can be configured with up to 256 standard current channels operating at:

- Up to 200mA with voltage limits of -2V to 7.2V
- Up to 20mA at voltage limits of -2V to 18V

Note that the UVS256 has specific current limits for sink and source fold levels. For more information, see Best Programming Practice When Choosing Fold Level and Current Range.

The UVS256 supports several configurations using multiple merging capabilities (with voltage limits of -2V to 2.5V), including:

- DCVS mode (unmerged) at up to 800mA (HC) or up to 200mA (SC)
- DCVSMerged2 mode (2 channels) at up to 1.4A (HC) or up to 400mA (SC)
- DCVSMerged4 mode (4 channels) at up to 2.8A (HC)
- DCVSMerged8 mode (8 channels) at up to 5.6A (HC)

This section includes the following topics:

- Features
- Safety Information

UVS256 information includes the following sections:

Related Topics

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About the UltraVS256 : Features

Features

The UVS256 is a highly parallel 256 channel device power supply (DPS). The UVS256 is compatible with the DCVS user programming model in IG-XL.

Each UVS256 instrument board has the following features:

- 256 instrument channels per slot
- Pattern control:
  - Vbump (Main/Alt Voltage DAC selection)
  - Trigger Capture Signal
  - Strobe Meter
  - Apply PSets
  - Filter On/Off
  - Alarm Enable/Disable
  - 63 PSets per channel

  - 16 Capture signals per channel
- PSet parameters:
  - HiZ/Force Voltage
  - CurrentRange
  - Source Fold Level
  - Sink Fold Levels
  - Meter Mode
  - Meter Filter
  - Alt Voltage
  - Main Voltage
  - Capture Sample Rate
  - Capture Sample Size
  - Bandwidth Setting
- DCVS programming instrument model
  - Standard IG-XL support, including: Pin Map, Channel Map, PSets, Signals, and Pin Levels sheets
  - Programming using DCVS VBT language, including: alarm, metering, PSet, capture signal, mode, and voltage/current related properties

  - Programming using DCVS patterns
  - Debugging using DCVS Debug Display
  - Programming using DCVSPowerSupply_T VBT Test
  - Standard autocalibration based on test program
  - Calibration and checkers through the Maintenance Environment
- Operating modes:
  - Force Voltage - Measure Voltage
  - Force Voltage - Measure Current
  - High Impedance - Measure Voltage
- DCVS (one channel) operating ranges:
  - High current channels: 800mA, 700mA, 200mA, 20mA, 200A, 20A, 4A
  - Standard current channels: 200mA, 20mA, 200A, 20A, 4A
- DCVSMerged2 (two channels merged) operating ranges:
  - High current channels: 1.4A, 700mA, 400mA, 200mA, 40mA, 20mA, 2mA, 200A, 20A, 4A
  - Standard current channels: 400mA, 200mA, 40mA, 20mA, 200A, 20A, 4A
- DCVSMerged4 (four channels merged) operating ranges:
  - High current channels: 2.8A, 700mA, 200mA, 20mA, 2mA, 200A, 20A, 4A

- DCVSMerged8 (eight channels merged) operating ranges:
  - High current channels: 5.6A, 700mA, 200mA, 20mA, 2mA, 200A, 20A, 4A
- Voltage limits:
  - -2V to 18V
    Current ranges supported: 4A, 20A, 200A, 2mA, 20mA
  - -2V to 7.2V
    Current ranges supported: 4A, 20A, 200A, 2mA, 20mA, 200mA
  - -2V to 2.5V
    Current ranges supported: 700mA, 800mA, 1.4A, 2.8A, 5.6A
- Programmable Current Clamps, support for:
  - Sink and Source Fold Levels

Note that source and sink fold levels have different minimum and maximum currents than the current range for the instrument. For more information, see Setting the Current Limit.

  - Shared Sink and Source Timeout
- Current and voltage meter measurement with hardware averaging support
  - Software go/no-go limit testing
- Current and voltage profiling with programmable capture sample rates and averaging support
- IIM/DSP support
- Programmable compensation using bandwidth settings of 0 to 255

For UVS256 hardware specifications, see UltraVS256 Specifications.

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About the UltraVS256 : Safety Information

Safety Information

As defined by IEC standards, an energy hazard exists when power levels exceed 240W. This type of hazard can produce sparks and cause conductive parts to weld to electrically conductive surfaces if they come in contact with the high-current sources. By this definition, the UVS256 board installed on an UltraFLEX test system represents an electrical hazard both to operators and equipment. Each voltage source on the UVS256 can supply a maximum of 2W (800mA at 2.5V). With 256 DCVS channels per UVS256 board, the total output power for this instrument is ~1700W.

For complete safety information, see About Operator and Safety Information.

Warning:

High current is available from the UVS256 instrument at the test head terminals. Follow the safety precautions listed below to avoid injuries due to electric shock or burns and to prevent damage to the equipment such as melting or burning of components and boards.

Observe the following safety precautions when using the UVS256.

- Advise operators of the potential energy hazard and ask them to exercise caution during testing, especially if four or more high-current voltage sources are operating in parallel.
- Do not touch the test head interface during testing or when the interface is powered on.
- Avoid using conductive tools when using the test system and do not wear rings, watches, bracelets, necklaces, or other objects that are metallic or conductive.

- Exercise caution when using conductive tools or probes. Make sure that these tools or probes do not come in contact with the high current sources or any other conductive part of the system.

- Exercise caution when probing on the test head to avoid inadvertent shorts between voltage or current terminals. Watch especially for situations where a probe could become loose and touch two voltage or current terminals.

- Wear an antistatic wrist strap with internal current limiting resistor because the UVS256 is subject to static discharge.

- Wear safety glasses.

- Exercise caution when designing and running tests and set appropriate current limits for tests. You may see large current changes by simply changing voltage levels or turning the large current-consuming elements of a device on and off. Verify the ripple current specifications for the DIB capacitors to meet these rapid current changing conditions. This is true any time the voltage is changing rapidly during Vbump, shmoo, and other testing activities.

- Keep the high-current areas clear of items that might alter their contacts. They must be clean and free of liquid or other conductors. Resistance from poor electrical contact may also result from physical damage, wear, or operation.

- Inspect the high-current area frequently and repair unsafe DIBs before use.

- Affix a “DANGER ELECTRICAL HAZARD” warning label to any DIB that uses the UVS256 instrument, including customer-designed DIBs.

- Make sure that DIBs using the UVS256 option have a protective cover with recessed or protected probe points, if present, to prevent operators from inadvertently touching the high-current components or dropping conductive tools or equipment onto them.

- Have an emergency response plan that all operators are familiar with. If an emergency occurs, immediately stop testing, get medical help if needed, and report the emergency to proper site personnel. Do not touch the test head as items may be burning, hot, or still electrically charged.

- If you detect smoldering, sparks, component burning or melting, or other severe failure, power down the test system immediately by pressing an EMO button. Do not touch any of the damaged parts as they may still be hot.

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About the UltraVS256 : Key Terms

Key Terms

This section uses the following terms, abbreviations and acronyms:

capture memory	The UVS256 hardware used to store sampled values. The UVS256 can capture up to 16K samples at a sample rate of 200kHz.
current profile	The plot of sampled current measurements over time.
DCVS	Direct Current Voltage Supply. This term is used to refer to each channel of the UVS256 board. Each DCVS on the UVS256 board is an instrument with independently programmable current limits, voltage and current meter, capture memory, strobe memory, and parameter set (PSet) memories.
DGS	Device ground sense. DGS is a sense signal that tracks the voltage of digital ground on the DIB. This is equivalent to other UltraFLEX boards' voltage sense ground (VSG).
force line	Wiring and DIB interposer that carry the voltage supply current to the DIB power planes and DUT.
PatGen	The UVS256 pattern generator can be used to synchronize UVS256 capabilities like capture, strobe, and Vbump to other instruments in the same PatGen domain. The PatGen controls the "real-time cadence of events" in a sequence.
PSet	A parameter set that sets the instrument state during a pattern burst. A PSet can also be applied in VBT.

sense line	An electrical connection used to measure the voltage at the Kelvin point. This connection is used to control the output of the force line to maintain the programmed output voltage at the desired level.
strobe memory	The UVS256 hardware used to store sampled strobe values. The UVS256 can capture up to 1K samples.
Vbump test	A test that changes the device power pin (VDD) level during operation to see if you can read from and write to the device correctly at different voltage levels. Vbump is also used to describe whenever the programmed voltage level for a device power supply is changed during a test pattern.

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About the UltraVS256 : UVS256 Software Licensing Requirements

UVS256 Software Licensing Requirements

The UVS256 does not require any licenses for the instrument. For general information on IG-XL licensing, see IG-XL Licensing.

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About the UltraVS256 : UVS256 Calibration

UVS256 Calibration

UVS256 calibration is handled internally by IG-XL and does not require the use of a calibration DIB.

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DCVS Debug Display

DCVS Debug Display

This section describes the DCVS instrument debug display used with the UVS256 instrument. This interactive DCVS debug display allows you to troubleshoot DCVS settings. You can change any values on the display and run your test program to see the effects of the change.

This section includes the following topics:

•	Starting the DCVS Debug Display
•	DCVS Debug Display Interface
•	Setting the Output Current Range and Meter Ranges
•	Disconnecting the Instrument

For more information on using debug displays, see the following:

•	Using Debug Displays in TDE
•	Producing Code from Debug Displays

Starting the DCVS Debug Display

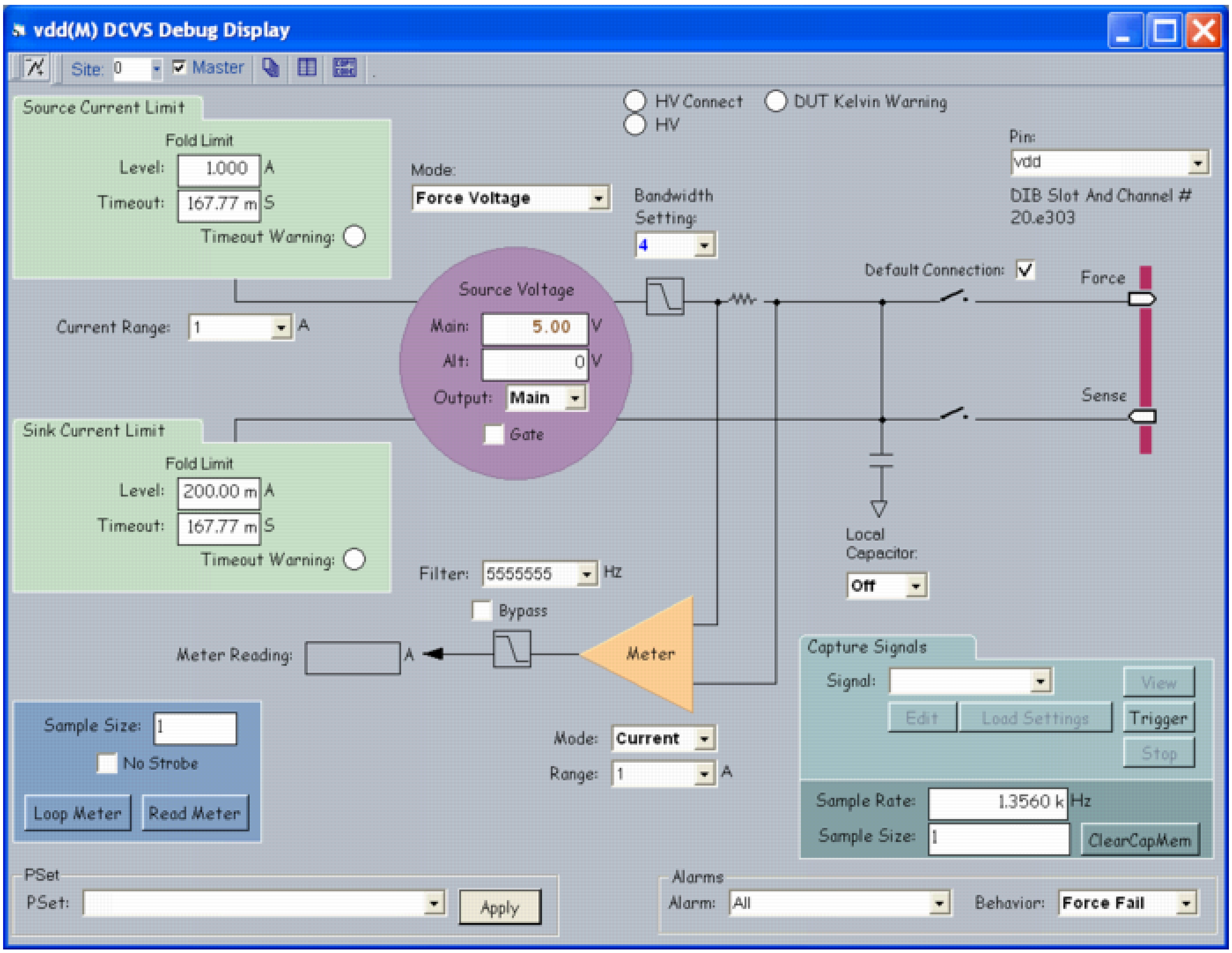
To start the DCVS debug display for UVS256, first validate an UVS256 test program, Then, on the IG-XL Tools tab, in the Launch group, select Debug Displays>DCVS. This opens the display for active sites only (see figure below).

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see Starting Debug Displays in the Test Debug Environment (TDE) section.

For information on the fields used in the DCVS debug display, see DCVS Debug Display Interface.



DCVS Debug Display with UVS256 Values

DCVS Debug Display Interface

This section describes the parameters, buttons, and fields used in the DCVS debug display. To change values in the debug display, either type the new values in the text box or select the new value from the menu.

For information on using debug displays and tools, see Test Debug Environment (TDE).

☑ **Note:**

To view the ranges or available options for fields that do not have drop-down menus, place your pointer over the field and the minimum and maximum values appear in a pop-up window. These ranges vary depending on which VS board you have assigned to the DCVS channel and how you have programmed it.

The table lists DCVS Debug Display parameters, buttons, and fields used by the UVS256.

DCVS Debug Display Fields

Menu, Button, or Field	Action
Source Current Limit	Use the following fields to view or set your source current limiting hardware: • Fold Limit: The fold current level limit in the Level field and the timeout period in the Timeout field. • Timeout Warning: Red circle indicates the fold timeout has expired. See Setting the Current Limit.
Sink Current Limit	Use the following fields to view or set your source current limiting hardware: • Fold Limit: The fold current level limit in the Level field and the timeout period in the Timeout field. • Timeout Warning: Red circle indicates the fold timeout has expired. See Setting the Current Limit.
Source Voltage	Use the following fields to set your output voltage: • Main: The main source voltage value in volts. • Alt: The alternate source voltage value in volts. • Output: The output for the source voltage, either Main or Alt.

	• Gate: The DCVS is gated on (checked) or off (empty). See Selecting VMain, VAlt, and IFoldLevel.
Mode	The DCVS operation mode. Options: • Force Voltage (On) • High Impedance (Off)
Pin	The pin you are observing. The DIB slot and channel number you are currently viewing are shown below this field.
Bandwidth Setting	The compensation mode. UVS256 values are 0 to 255. See Setting the Output Compensation Bandwidth.
Current Range	The output current range you are forcing. See Setting the Output Current Range and Meter Ranges.
Default Connection	Checkbox determines how the force and sense connections are opened and closed. When checked, these connections are opened and closed together. For UVS256, this checkbox is always checked, since UVS256 doesn't support connect/disconnect force and sense individually.
Local Capacitor	Not used in UVS256. The connection state of the local output capacitor. When the DCVS is disconnected from the DIB, the default capacitor keeps the DCVS voltage output from oscillating. Options: • On • Off • Auto UVS256 doesn't support local capacitor and its state is always off. The only valid setting is Off.
Meter	The DCVS meter settings, including:

	• Mode: The meter mode, either Current or Voltage. • Range: Not used in UVS256. The meter range. • Filter: The meter bandwidth (for viewing only). • Bypass: Checking this box disables filtering, allowing you to view a higher frequency signal with more stability. • Meter Reading: The measured voltage or current value. • Sample Size: The number of samples strobed. • No Strobe: Checking this box disables the strobe function. • Loop Meter: Clicking this button opens the Meter Loop Tool and causes the meter to loop through multiple strobe sets. See Meter Loop Tool. • Read Meter: Clicking this button prints the average of the strobed samples. See Setting the DCVS Current Meter.
PSet	A list of defined PSets for selected pin site. When no PSet is defined, the list is empty. When a PSet is selected, the PSet is applied to instrument hardware immediately. The box is updated to display any PSet that is applied from VBT.
Alarms	The alarm type and behavior for the selected alarm. • Alarm: The alarm type. • Behavior: The action that the alarm takes when triggered. See Alarms Subsystem and Setting Alarms.

Capture Signals	The capture memory signal settings, including: • Signal: A capture signal tag listed in the drop-down menu. • Load Settings: Loads the capture signal tag's settings. • Edit: Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. See PSet and Signals Editor. • View: Shows the selected capture signal tag's waveform. • Trigger: Triggers new capture signal with the current hardware setup. • Stop: Stops the current capture, if one is running. • Sample Rate: The capture rate currently set in hardware. • Sample Size: The capture sample size currently set in hardware. • ClearCapMem: Clicking this button clears both the signal (capture) and meter memories for channels behind the specified pin or pins. See Setting Up DCVS Capture.
DUT Kelvin Warning	A red circle indicates that the DUT Kelvin threshold has been exceeded.
HV Gate	A red circle indicates that the high voltage gate (HV_GATE) safety signal from the support board is not connected properly to DIB ground. This usually means that the DIB is either not properly seated or designed incorrectly.

HV Connect	A red circle indicates that the high voltage connect (HV_CONNECT) safety signal from the support board is not connected properly to DIB ground. This usually means that the DIB is either not properly seated or designed incorrectly.

Setting the Output Current Range and Meter Ranges

The available output current ranges and corresponding meter ranges depend on whether you have merged your DCVS channels or left them independent. For more information, see the table in UVS256 Merging.

Disconnecting the Instrument

If you disconnect the UVS256 instrument, it goes into a disconnect state and will not respond to any changes you make using the display. Once you reconnect the instrument, any changes you made will be applied in a specific order. For more information, see Instrument Disconnect State.

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UltraVS256 Examples

UltraVS256 Examples

This section contains two kinds of program examples for the UVS256:

- IG-XL workbook examples show all IG-XL data, VBT code, and patterns (if used) necessary for a complete running program.
- VBT function examples show one or more VBT functions that perform a programmed action when called as specified.

Workbook Examples

- UVS256 Static Icc Programming

VBT Function Examples

- Basic DCVS Setup Programming
- Vbump Testing
- Current Profiling

Related Topics

- DCVS (VBT Syntax Reference)

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UltraVS256 Examples : UVS256 Static Icc Programming

UVS256 Static Icc Programming

This workbook example shows how to program an UVS256 static Icc test that forces voltage and measures current (FVMI) on two SN74ALS245AN bidirectional digital transceiver devices on two sites. This example connects one UVS256 channel to the Vcc pin, applies 5.5V to the pin, and measures the resulting current while the digital pins are set to a known state. This example uses one HSD1000 board in slot 16 and an UVS256 board in slot 22.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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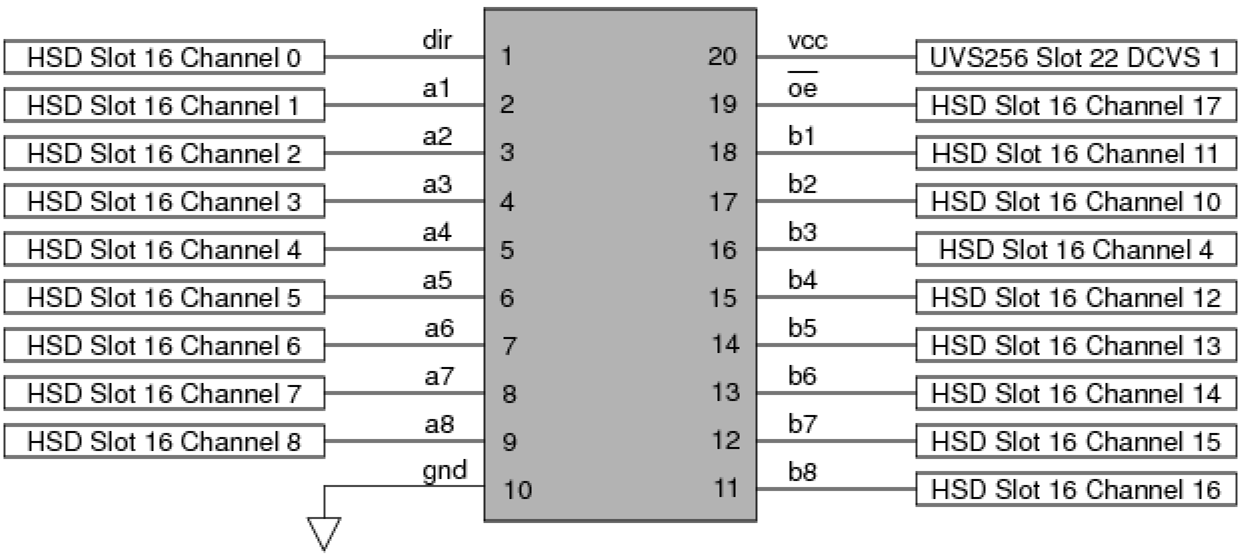
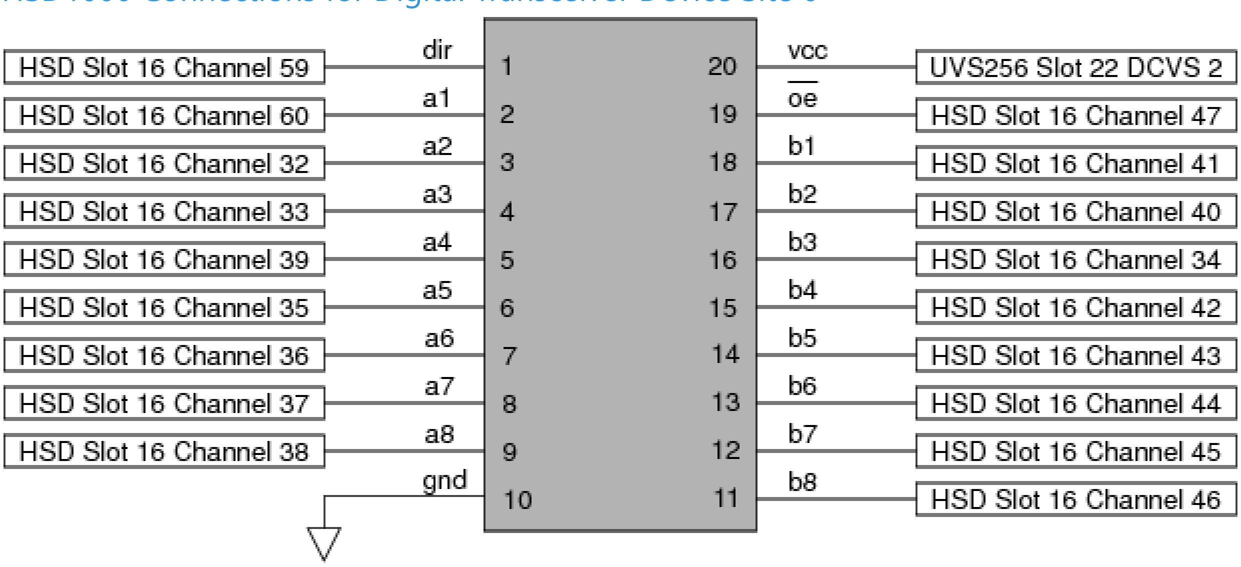
UltraVS256 Examples : UVS256 Static Icc Programming : DUT Connections

DUT Connections



HSD1000 Connections for Digital Transceiver Device Site 0



HSD1000 Connections for Digital Transceiver Device Site 1

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UltraVS256 Examples : UVS256 Static Icc Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Pin Levels Sheet
- Global Specs Sheet
- Test Instances Sheet
- Flow Table Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	Gnd
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc	Power
ctrls	oe	Input
ctrls	dir	
porta	a1	I/O
portb	b1	I/O
ports	porta	I/O
ports	portb	
digital_pins	ctrls	I/O
digital_pins	ports	

porta groups pins a1-a8.
portb groups pins b1-b8.

Channel Map Sheet

Channel Map

DIB ID: View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc	Pin 20	DCVS	22.sense1	22.sense2

Pin Levels Sheet

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc	0	Vps	5.5
vcc		Isc	=_gl_Isc
vcc		Tdelay	0.1
ctrls		Vil	0
ctrls		Vih	5
ctrls		Vt	2.5
ctrls		Vcl	=_Vcl_default
ctrls		Vch	=_Vch_default
ctrls		DriverMode	Largeswing-HiZ
ports		Vil	0
ports		Vih	5
ports		Vol	0.8
ports		Voh	2.4
ports		Vt	=0
ports		Vcl	=_Vcl_default
ports		Vch	=_Vch_default
ports		VoutLoTyp	0.1
ports		VoutHiTyp	4.9
ports		DriverMode	Largeswing-VT1K

Values shown in formula.

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc	0	Vps	5.5
vcc		Isc	1
vcc		Tdelay	0.1
ctrls		Vil	0
ctrls		Vih	5
ctrls		Vt	2.5
ctrls		Vcl	-1
ctrls		Vch	6
ctrls		DriverMode	Largeswing-HiZ
ports		Vil	0
ports		Vih	5
ports		Vol	0.8
ports		Voh	2.4
ports		Vt	0
ports		Vcl	-1
ports		Vch	6
ports		VoutLoTyp	0.1
ports		VoutHiTyp	4.9
ports		DriverMode	Largeswing-VT1K

Values shown after evaluation.

Global Specs Sheet

Global Specs

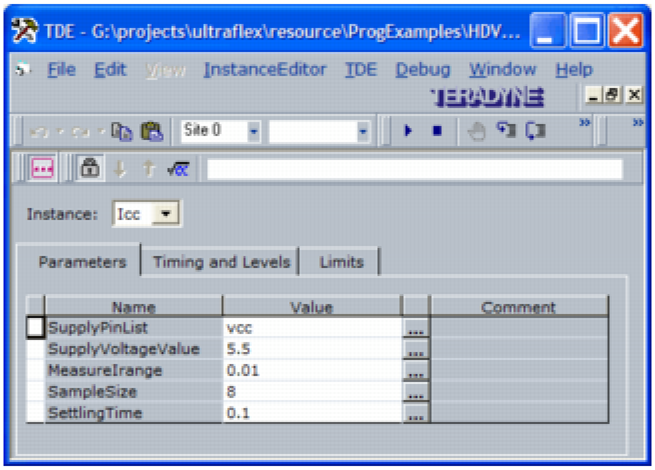
Symbol	Job	Value	Comment
Vcl_default		-1	Detector clamp voltage low
Vch_default		6	Detector clamp voltage high
gl_Isc		1	
gl_Tdelay		=20*ms	

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
Icc	VBT	Icc_VBT								Levels		

The Icc test instance has the following parameters and values as shown in the Instance Editor:



Flow Table Sheet

Flow Table

		Gate			Command				Limits		Bin Number		Sort Number		
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Pass	Fail	Pass	Fail	Result
					set-error-bin							99		999	
					Test	Icc									
					Use-Limit	Icc			0.025	0.085		4		40	Fail
					set-device						1		1		Pass

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UltraVS256 Examples : UVS256 Static Icc Programming : Patterns

Patterns

This program example does not require any pattern files.

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UltraVS256 Examples : UVS256 Static Icc Programming : VBT Code

VBT Code

```
Option Explicit
Public Function Icc_VBT(SupplyPinList As PinList, _
                        SupplyVoltageValue As Double, _
                        MeasureIrange As Double, _
                        SampleSize As Long, _
                        SettlingTime As Double) As Long
    Dim Measure As New PinListData
    On Error GoTo errHandler
    ' Apply HSD levels and power supply pin values.
    TheHdw.Digital.ApplyLevelsTiming True, True, False, tlPowered
    With TheHdw.DCVS.Pins(SupplyPinList)
        ' Override actual programmed SupplyVoltageValue if required.
        If SupplyVoltageValue <> 0 Then
            ' Use specified SupplyVoltageValue for test.
            .Voltage.Main.Value = SupplyVoltageValue
        Else
            ' Read back actual programmed force level value for datalog.
            SupplyVoltageValue = .Voltage.Main.Value
        End If
        ' - - - Set up the DCVS instrument to perform Icc measurement. - - -
        ' The current limit is set within the ApplyLevelsTiming statement.
        ' (LoadLevels = True)
        .Meter.Mode = tlDCVSMeterCurrent
        .Meter.CurrentRange.Value = MeasureIrange
```

```
        ' Use UVS256 hardware averaging.
        .Meter.Filter.Value = .Meter.Filter.Max / SampleSize
        TheHdw.Wait (SettlingTime)
        ' Strobe meter on the SupplyPinList pins
        Measure = .Meter.Read(tlStrobe, 1) ' Due to UVS256 HW averaging, just
                                           ' one sample must be read back.
    End With
    ' - - - Datalog Results - - -
    TheExec.Flow.TestLimit Measure, , , tlSignGreaterEqual, _
                tlSignLessEqual, scaleNone, unitAmp, , _
                TheExec.DataManager.instanceName, _
                CompareAverage, SupplyPinList, _
                SupplyVoltageValue, unitVolt, , , tlForceFlow
    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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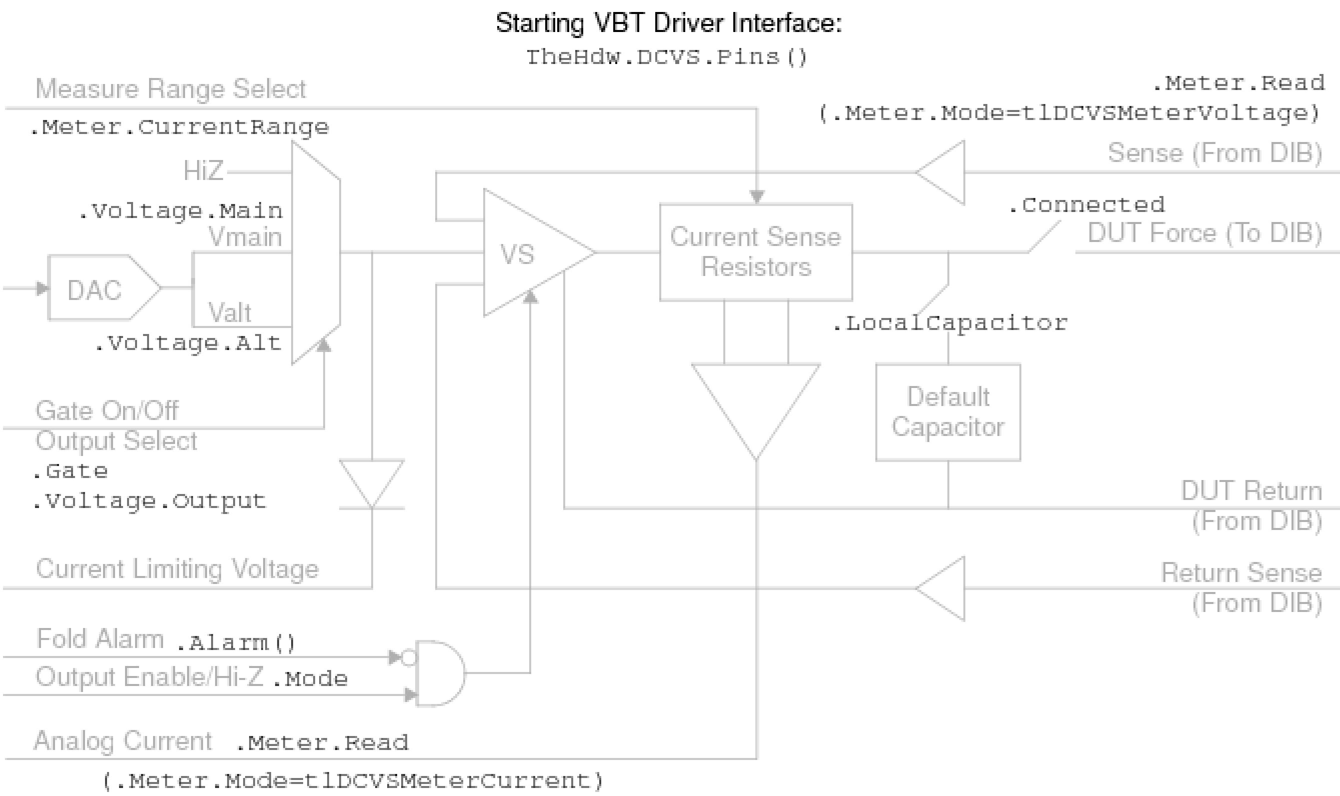
UltraVS256 Examples : Basic DCVS Setup Programming

Basic DCVS Setup Programming

The following example describes how to program the main output subsystem as shown in the following figure.



DCVS Instrument: Programming the Main Output Subsystem

```
Sub DCVS_BasicSetup()
  With TheHdw.DCVS.Pins("VDD")
  ' Set the DCVS bandwidth to optimize for desired transient response.
  .BandwidthSetting = 3
  ' You can force merged DCVS into a lower current output mode. The lower
  ' current output mode is necessary for tighter current limiting
```

```
    ' specifications and low current metering.
    .CurrentRange.Value = .CurrentRange.Min
    ' Set the main and alt voltages
    .Voltage.Main.Value = 2.5
    .Voltage.Alt.Value = 2.1
    ' Set the output to Main DAC value. You can select Alt later using VBT or a
    ' pattern opcode
    .Voltage.Output = tlDCVSVoltageMain
    ' Set the DCVS output mode to Voltage (not High Impedance)
    .Mode = tlDCVSModeVoltage
    ' Connect the DCVS output (This connects both force and sense)
    .Connect (tlDCVSConnectDefault)
    ' Gate on the DCVS
    .Gate = True
    End With
End Sub
```

You can use the following code to verify your force and sense connections.

```
With TheHdw.DCVS.Pins(...)
    If .Connected = tlDCVSConnectForce Then
        Debug.Print "Force Connected"
    End If
    If .Connected = tlDCVSConnectSense Then
        Debug.Print "Sense Connected"
    End If
    If .Connected = tlDCVSConnectSense + tlDCVSConnectForce Then
        Debug.Print "Force and Sense Connected"
    End If
    If .Connected = tlDCVSConnectNone
        Debug.Print "No Connection"
    End If
End With
```

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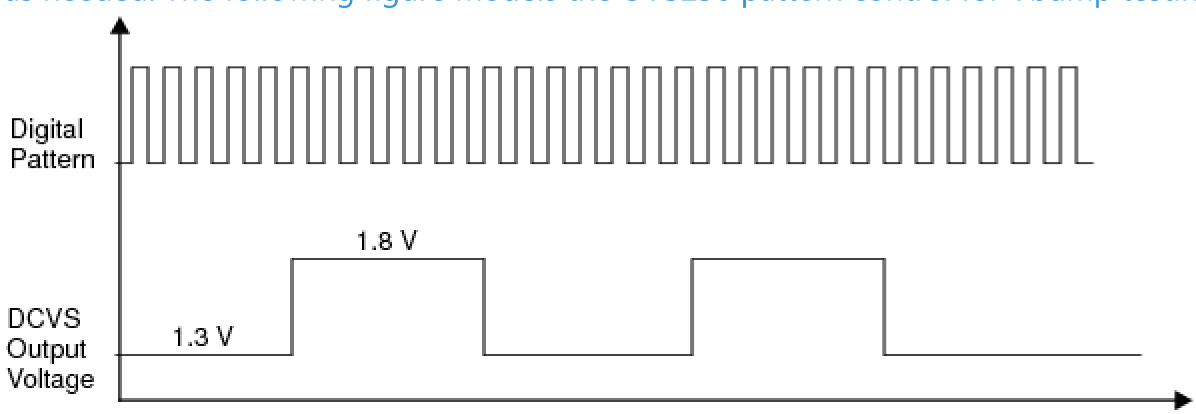
UltraVS256 Examples : Vbump Testing

Vbump Testing

A Vbump test changes the device power pin (VDD) level during a test pattern burst. You may need to conduct a Vbump test to verify that a device is making smooth power transitions as it moves between different power states. For example, you could check a device's operation at two different voltage levels (1.3V and 1.8V).

To set up a Vbump test, the UVS256 can change the voltage during a burst with Vmain, Valt, and parameter set programming. Note that the UVS256 also uses the high-speed current meter to collect the Vbump test data during pattern bursts.

To give you pattern control for Vbump testing, the UVS256 allows you to select one of two voltage levels during pattern bursts. Within a pattern burst, you can start and stop the current meter only as needed. The following figure models the UVS256 pattern control for Vbump testing.



DCVS Voltage Pattern Control for Vbump Testing

Note that bumping the voltage level may cause current clamping and may trigger an alarm. In such cases, you can use the following VBT syntax to turn off alarms.

TheHdw.DCVS.Pins(Pinlist).Alarm(AlarmType) = tlAlarmOff

When turning off alarms to avoid expected alarms, remember to turn them back on again as soon as possible.

For more information, see Setting Alarms.

Vbump Example Using VBT PSets and a Pattern File

The following example uses PSets and a pattern file with the VS_Alt_Sel and VS_Main_Sel microcodes to “bump” the output voltage on a pin named VDD between four different values in the following order:

1.	Start at 3.0V.
2.	Vbump to 3.5V.
3.	Vbump to 3.0V.
4.	Vbump to 1.5V.
5.	Vbump to 1.75V.
6.	Vbump to 1.5V.
7.	End at 1.5V.

VBT PSet Programming Function

This example calls the AddPset function to create a PSet named "my_pset_1". This function is described in the DCVS PSet Example Function. For more information on programming with PSets, see PSets.

```
Function VBumpSetup(argc As Long, argv() As String) As Long
    ' Set voltage levels and define the PSet before pattern burst
    ' (For example, PrePat interpose function)
    Dim power_supply_pin As String
    ' Set the power supply pin to VDD
    power_supply_pin = "VDD"
    TheHdw.DCVS.Pins(power_supply_pin).Voltage.Main.Value = 3
    TheHdw.DCVS.Pins(power_supply_pin).Voltage.Alt.Value = 3.5
    ' Create "my_pset_1" using the AddPSet() function.
    Call AddPSet(power_supply_pin, _
        PSetName:="my_pset_1", _
        BandwidthSetting:=255, _
```

```
        CurrentRange:=1, _
        SampleRate:=5555555, _
        SampleSize:=500, _
        SourceFoldLimit:=0.7, _
        MeterCurrentRange:=1, _
        MeterVoltageRange:=7, _
        MeterMode:=tlDCVSMeterCurrent, _
        MainVoltage:=1.5, _
        AltVoltage:=1.75)
    VBumpSetup = TL_SUCCESS
End Function
```

Pattern File Example

The embedded comments in the following pattern file show where Vbump changes occur. For more information on programming pattern files, see DCVS Pattern Control. For settle time guidelines, see DCVS Pattern File Example.

```
import tset ts1;

instruments = {
    VDD : DCVS;
}
srm_vector ($tset, dig1:S)
{
    // ___ THIS PATTERN STARTS WITH VDD VOLTAGE at 3.0V  ___
    > ts1   1;
    > ts1   1;
    //... many more vectors
    > ts1   1;
    > ts1   1;
    // ___ VBUMP to 3.5 V ___
    (vdd : DCVS = VS_Alt_Sel)
    > ts1   1;
    // settle time repeat. See guidelines in "DCVS Pattern File Example."
    repeat 5000
    > ts1   1;
    > ts1   1;
    //... many more vectors
    > ts1   1;
    > ts1   1;
```

```
// ___ VBUMP TO 3.0 V ___
(vdd : DCVS = VS_Main_Sel)
> ts1   1;
// settle time repeat. See guidelines in "DCVS Pattern File Example."
repeat 5000
> ts1   1;
> ts1   1;
//... many more vectors
> ts1   1;
// ___ APPLY A PSET TO LOAD NEW VALUES OF MAIN AND ALT ___
// ___ THIS PSET CHANGES VDD TO 1.5 V ___
(vdd : DCVS = PSet my_pset_1)
> ts1   1;
> ts1   1;
//... many more vectors
> ts1   1;
> ts1   1;
//... many more vectors
> ts1   1;
> ts1   1;
// ___ VBUMP TO 1.75 V ___
(vdd : DCVS = VS_Alt_Sel)
> ts1   1;
// settle time repeat. See guidelines in "DCVS Pattern File Example."
repeat 5000
> ts1   1;
> ts1   1;
//... many more vectors
> ts1   1;
> ts1   1;
// ___ VBUMP TO 1.5 V ___
(vdd : DCVS = VS_Main_Sel)
> ts1   1;
// settle time repeat. See guidelines in "DCVS Pattern File Example."
repeat 5000
> ts1   1;
> ts1   1;
> ts1   1;
```

```
    //... many more vectors
    > ts1   1;
}
```

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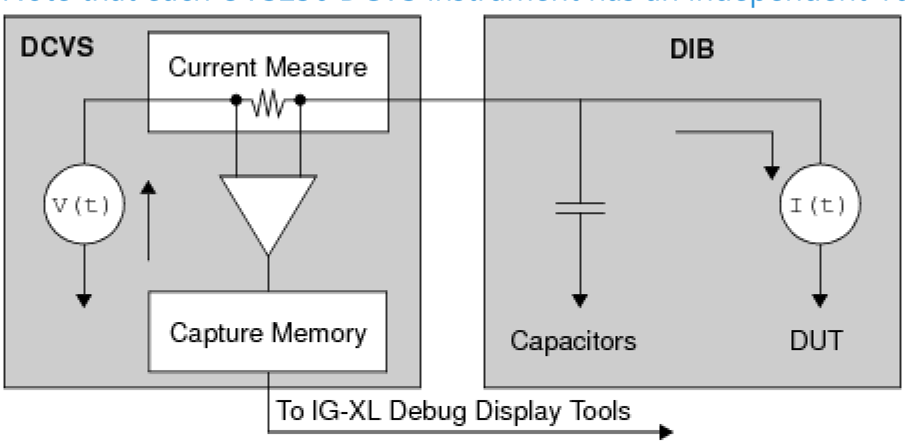
UltraVS256 Examples : Current Profiling

Current Profiling

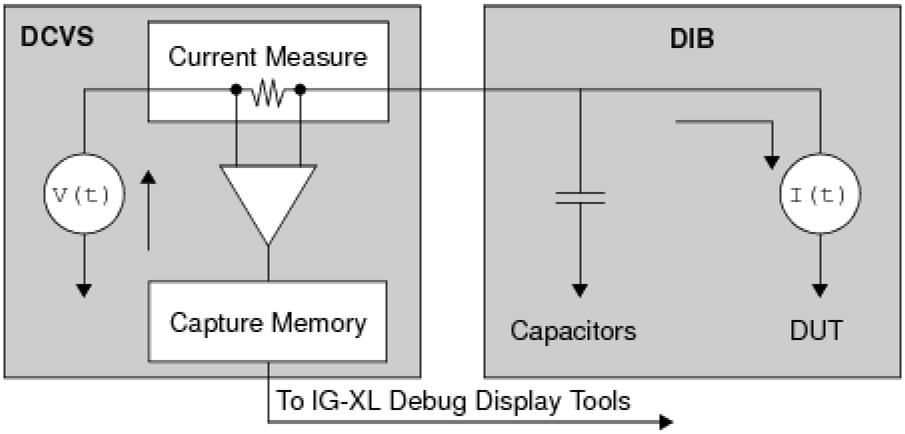
A current profile is a collection of data points of a device's current draw over time. Generally, you conduct a current profile to see what a tested part is doing as you cause it to pull more or less current. This allows you to monitor the current profile during the transition between device power states. For instance, you may want to verify that the device is drawing the expected current and is not experiencing current spikes as it moves between inactive and active modes.

The UVS256 also allows you to collect a current profile with a single, continuous, full-speed PatGen burst. The data is sampled at the specified frequency and stored in the on-board capture memory. Note that each UVS256 DCVS instrument has an independent 16K capture memory.



Current Profile Circuit Diagram

The figure below shows a hypothetical current profile generated from multiple test patterns run at speed to test a device's current requirements as it moves through different power states. The current changes as the device goes through the power-up sequence, the phase-locked loop (PLL) warm-up, and its normal and sleep modes.



Current Profile Sample

Capture Setup, Trigger, and Plot Programming

The following example shows how to capture a current or voltage waveform and plot the result.

```
Sub subProfile_Start(PlotName As String, _
              PinName As String, _
              Optional VorI As tlDCVSMeterMode = tlDCVSMeterCurrent, _
              Optional SampleRate As Double = 100000, _
              Optional SampleSize As Long = 500)
  Dim TestCaptureLabel As String
  If SampleRate < TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Min Then
    SampleRate = TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Min
  ElseIf SampleRate > TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Max Then
    SampleRate = TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Max
  End If
  If SampleSize < TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Min Then
    SampleSize = TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Min
  ElseIf SampleSize > TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Max Then
    SampleSize = TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Max
  End If
  TheHdw.DCVS.Pins(PinName).Meter.Mode = VorI
  TestCaptureLabel = PlotName
  ' Wait if a capture is already running.
  Do While TheHdw.DCVS.Pins(PinName).Capture.IsRunning = True
  Loop
  TheHdw.DCVS.Pins(PinName).Capture.Signals.Add TestCaptureLabel
  TheHdw.DCVS.Pins(PinName).Capture.Signals.DefaultSignal = TestCaptureLabel
  With TheHdw.DCVS.Pins(PinName).Capture.Signals.Item(TestCaptureLabel)
    .Reinitialize
    .Mode.Mode = tlSignalModeUseLoadedValue
```

```
    ' When calling .LoadSettings, do not override the .Meter.Mode.
    ' ____ NOTE on Signals.Item(<label>).<signal_parameter>.Mode ____
    ' The signal parameter mode is a LoadSettings Mode for the signal.
    ' It determines if the parameter is set to the Signal-defined value
    ' or remains untouched when the .LoadSettings method is called.
    ' This example uses the .Mode.Mode syntax, which determines the
    ' behavior of the .LoadSettings function for the parameter. In the
    ' example syntax above, .Mode.Mode, the first .Mode is the
    ' DCVS.Meter.Mode (voltage or current). The second .Mode is the signal
    ' LoadSettings mode. All signal parameters have a LoadSettings mode.
    ' For instance you can also program:
    ' .SampleRate.Mode = tlSignalModeUseLoadedValue ' Use value already loaded to hardware.
Do not override hardware.
    ' .SampleSize.Mode = tlSignalModeUseLoadedValue ' Use value already loaded to hardware. Do
not override hardware.
    ' The default value is tlSignalModeUseValue (Use the value defined for the signal; that is,
override current hardware.)
    .SampleRate = SampleRate
    .SampleSize = SampleSize
    .LoadSettings
    .Trigger
  End With
End Sub
Sub subProfile_Plot(PlotName As String, _
             PinName As String, _
             Optional DumpToFile As Boolean = False)
  Dim TestCaptureLabel As String
  Dim DSPW As New DSPWave
  ' Wait if capture is still running.
  Do While TheHdw.DCVS.Pins(PinName).Capture.IsRunning = True
  Loop
  TestCaptureLabel = PlotName
  Dim FileName As String
  Dim strMsg As String
  ' Single-pin PinListData can be read directly into DSPWave
  DSPW = TheHdw.DCVS.Pins(PinName).Capture.Signals.(TestCaptureLabel).DSPWave
  If DSPW.SampleSize > 0 Then
    DSPW.Plot TestCaptureLabel
```

```
    If DumpToFile Then
      FileName = "PlotData_" & TestCaptureLabel & ".txt"
      Call PlotToFile(FileName, DSPW)
    End If
  Else
    strMsg = "subProfile_Plot() Warning: No Samples Captured (CaptureLabel=" &
TestCaptureLabel & ")"
    TheExec.Datalog.WriteComment (strMsg)
    Beep
  End If
  Set DSPW = Nothing
End Sub
Sub PlotToFile(PlotFilePathAndName As String, PlotDSPW As DSPWave)
  Dim SamplePeriod As Double
  Dim SampleSize As Long
  Dim s As Long
  Dim FileNo As Integer
  Dim TempArray() As Double
  FileNo = 7
  Open PlotFilePathAndName For Output As #FileNo
  SampleSize = PlotDSPW.SampleSize
  SamplePeriod = (1 / PlotDSPW.SampleRate)
  TempArray = PlotDSPW.data
  For s = 0 To (SampleSize - 1)
    Print #FileNo, s * SamplePeriod & ", " & TempArray(s)
  Next s
  Erase TempArray
  Close #FileNo
End Sub
```

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UltraVS256 DCVS Programming

UltraVS256 DCVS Programming

This section describes programming of UVS256 instruments.

The following topics are available:

- Simulated Configuration Files
- Programming Procedure Overview
- Before You Begin
- DCVS DataTool Programming
- DCVS VBT Programming
- DCVS Pattern Control
- Using UVS256 in Concurrent Testing

Related Topics

- DCVS (VBT Syntax Reference)
- UltraVS256 Examples

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UltraVS256 DCVS Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the UVS256 offline by forcing IG-XL to map the UVS256. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the UVS256 board in slot 22. Note that the UVS256 board is designated as "VHDVS" in the config file.

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type              idprom (type serial rev company)

0.0            HSD-4G           974-245-10 900c56f 0905-A 5445
               HSD-4G           956-128-02 304c364 0846-B 5445
               PowerBoard        956-462-10 304b857 0741-A 5445
               ChanBoard0        956-160-10 3044ae4 0905-A 5445
               ChanBoard1        956-160-10 3044ad4 0905-A 5445
2.0            HSD-4G           974-245-10 900c570 0905-A 5445
```

HSD-4G 956-128-02 3044b11 0846-B 5445
PowerBoard 956-462-10 304b85c 0905-A 5445
ChanBoard0 956-160-10 3044ad2 0905-A 5445
ChanBoard1 956-160-10 3044adf 0905-A 5445
20.0 HDVS1 974-232-00 03036816 0649-B 5445
HDVS1 956-071-00 03025DF9 0649-B 5445
HDVS1Power 956-073-00 0303481D 0630-B 5445
20.1 HDVS1Rider-0 974-233-00 03111297 0634-B 5445
HDVS1Rider-0 956-072-00 03111297 0634-B 5445
20.2 HDVS1Rider-1 974-233-00 031112A6 0634-B 5445
HDVS1Rider-1 956-072-00 031112A6 0634-B 5445
20.3 HDVS1Rider-2 974-233-00 0311129D 0634-B 5445
HDVS1Rider-2 956-072-00 0311129D 0634-B 5445
20.4 HDVS1Rider-3 974-233-00 0311129F 0634-B 5445
HDVS1Rider-3 956-072-00 0311129F 0634-B 5445
20.5 HDVS1Rider-4 974-233-00 031112B2 0634-B 5445
HDVS1Rider-4 956-072-00 031112B2 0634-B 5445
20.6 HDVS1Rider-5 974-233-00 03111298 0634-B 5445
HDVS1Rider-5 956-072-00 03111298 0634-B 5445
20.7 HDVS1Rider-6 974-233-00 03111295 0634-B 5445
HDVS1Rider-6 956-072-00 03111295 0634-B 5445
20.8 HDVS1Rider-7 974-233-00 0311129E 0634-B 5445
HDVS1Rider-7 956-072-00 0311129E 0634-B 5445
20.9 HDVS1Rider-8 974-233-00 031112A0 0634-B 5445
HDVS1Rider-8 956-072-00 031112A0 0634-B 5445
20.10 HDVS1Rider-9 974-233-00 031112B0 0634-B 5445
HDVS1Rider-9 956-072-00 031112B0 0634-B 5445
20.11 HDVS1Rider-10 974-233-00 031112A3 0634-B 5445
HDVS1Rider-10 956-072-00 031112A3 0634-B 5445
20.12 HDVS1Rider-11 974-233-00 031112B3 0634-B 5445
HDVS1Rider-11 956-072-00 031112B3 0634-B 5445
22.0 VHDVS 805-052-00 0304B8D1 0902-A 5445
VHDVS 939-420-00 0304B89D 0845-A 5445
22.1 VHDVSPower 805-054-00 0304C509 0847-A 5445
VHDVSPower 939-422-00 0304C509 0847-A 5445
22.2 VHDVSRider-1 805-053-00 030350E3 0902-A 5445
VHDVSRider-1 939-421-01 030350E3 0902-A 5445
22.3 VHDVSRider-0 805-053-00 030350E4 0902-A 5445

	VHDVSRider-0	939-421-01 030350E4 0902-A 5445
24.0	SupportBoard	974-221-03 c016e45 0744-A 5445
	SupportBoard	956-416-30 c01838c 0739-B 5445
	PowerBoard	956-359-10 c0183aa 0744-C 5445
	TCM	956-419-02 c0183be 0637-C 5445
	DCCM	949-938-02 C0183D0 0744-C 5445
62.0	DistributionBoard	956-455-00 3036a4f 0723-A 5445
	DistributionBoard	956-455-00 3036a4f 0723-A 5445

To use the UVS256 offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the UVS256 online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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UltraVS256 DCVS Programming : Programming Procedure Overview

Programming Procedure Overview

To develop a test program that uses the DCVS instrument, follow this basic procedure:

1. Determine device power supply requirements (Device Power Supply Requirements).
   a. Number of power pins
   b. Output voltage and current ranges for each power pin
   c. Maximum current limiting requirements
2. Consider the following features specific to the DCVS instrument (DCVS Instrument Considerations):
   a. Instrument board-specific DIB design considerations
   b. Output power specifications
   c. Merging DCVS instrument supplies
3. Create a Pin Map sheet (Programming the Pin Map).
4. Create a Channel Map sheet (Programming the Channel Map).

5.	Create a Pin Levels sheet (Programming Pin Levels).
6.	Set the maximum current limits using VBT code (Setting the Current Limit).
7.	Set the compensation mode using VBT code (Setting the Output Compensation Bandwidth).
8.	Set the alarms using VBT code (Setting Alarms).
9.	Set the DCVS meter to measure current using VBT code (Setting the DCVS Current Meter).
10.	Create DCVS test patterns to use in your test instances (DCVS Pattern Control).
11.	Create test instances in the Test Instances sheet (DCVS Test Instance Programming).
12.	Add test instances to the Flow Table sheet.
13.	Run the test program, debug run-time errors or alarms, and optimize for run-time performance.

VBT syntax is described in the DCVS section of the VBT Syntax Reference.

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UltraVS256 DCVS Programming : Before You Begin

Before You Begin

Before writing a DCVS test program, you must determine the power requirements for your device application, including the number of power pins, the output voltage and current ranges for each power pin, and the maximum current limiting requirements. Moreover, you must understand the features of the DCVS instrument and how they affect your test program development. This section covers both of these topics:

- Device Power Supply Requirements
- DCVS Instrument Considerations

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UltraVS256 DCVS Programming : Before You Begin : Device Power Supply Requirements

Device Power Supply Requirements

Review the device datasheet to determine the device's power pin requirements. Note, however, that during testing, the device power requirements are likely to exceed the standard functional values listed in the datasheet. When possible, therefore, consult the device product engineer to get an idea of the predicted device power requirements during testing. In general, new generation digital VLSI devices tend to draw more power supply current during testing than previous generation devices.

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UltraVS256 DCVS Programming : Before You Begin : DCVS Instrument Considerations

DCVS Instrument Considerations

Consider the following before creating a DCVS test program.

DCVS Instrument DIB Design Guidelines

For information on DIB design guidelines, including DIB signals and connections and wiring guidelines, see UltraVS256 DIB Design Guidelines.

UVS256 High Current (HC) Channels

Standard current (SC) channels cannot access ranges above 200mA for a single channel or above 400mA for two merged channels. The UVS256 high current (HC) feature provides additional high current ranges:

- 700mA and 800mA ranges on non-merged channels
- 700mA and 1400mA ranges on DCVSMerged2 channels
- 700mA and 2800mA ranges on DCVSMerged4 channels
- 700mA and 5600mA ranges on DCVSMerged8 channels

The DCVS programming model is extended to support this feature.

A channel is high current channel when the signal name or DIB Pogo string with the “hc” suffix is used in a channel map. A channel is a standard current channel when the signal name or DIB Pogo string without the “hc” suffix is used.

An IG-XL pin that is assigned to a high current channel is high current pin. The same definition applies for a standard current pin. Unmerged and DCVSMerged2 channels can be defined as high current or standard current channels in a channel map. DCVSMerged4 and DCVSMerged8 channels are automatically high current channels. UVS256 does not support mixing high current and standard current channels across sites for an IG-XL pin in a channel map. All sites for a pin must either be assigned to high current channels or standard current channels.

Each UVS256 board has two channel boards, with 128 channels on each board. You can designate any 64 channels within a channel board as high current channels. An unmerged high current channel is counted as one high current channel. A DCVSMerged2 high current channel is counted as two high current channels: one high current leader channel and one high current follower channel. The same counting rule applies for DCVSMerged4 high current and DCVSMerged8 high current channels.

☑ **Note:**

Do not mix UVS256 HC and SC channels in a pin group. IG-XL returns a validation error for this invalid programming setup.

Signal Delivery Path Current Limitation

The signal delivery path on the UVS256 is limited to 2.8A. Due this limitation, the sum of source fold levels within a predefined group of four channels cannot exceed 2.8A. The predefined groups of four channels with the 2.8A limitation on an UVS256 board are:

{ch1, ch2, ch3, ch4}, {ch5, ch6, ch7, ch8},..., {ch253, ch254, ch255, ch256}

Keep this 2.8A limitation in mind in the following situations:

- When using all four channels as unmerged high current channels
- When using the combination of two unmerged high current channels and a DCVSMerged2 high current channel

Channel Assignment Guidelines

The following are guidelines to consider when assigning UVS256 channels:

- If a pin needs more than 200mA, use an HC channel.
- If a pin needs more than 800mA, consider merging channels.
- If a pin needs more than 5.6A, consider using another instrument than the UVS256.
- If a pin needs more than 200mA but less than 800mA, and you are out of HC channels, consult Teradyne Applications.

- Start with "highest merge" pins, then do next highest, and so on.
- Spread HC channels out across entire instrument, leaving gaps. In the gaps, spread SC channels out across the entire instrument.
- If possible, assign channels n...n+3 (where n = 1, 5, 9, etc.) to the same DUT. This arrangement is helpful because sets of four channels on the UVS256 share the same Low Sense.
- Do not mix UVS256 HC and SC channels in a pin group. IG-XL returns a validation error for this invalid programming setup.

UVS256 Merging
UVS256 supports:

- Merging of two channels (DCVSMerged2 channel)
- Merging of four channels (DCVSMerged4 channel)
- Merging of eight channels (DCVSMerged8 channel)

The DCVS programming model for UVS256 supports the DCVS, DCVSMerged2, and DCVSMerged4, and DCVSMerged8 channel types in an IG-XL channel map. (UVS256 does not support the DCVSMerged6 channel type.)
Only the leader channel is used to define a merged channel in a channel map. See the table below for the dedicated leaders for each merged configuration.
Also note the following:

- UVS256 does not support merging channels from different boards.
- The accuracy specification for unmerged channels differs from the specification for merged channels.

The following table lists UVS256 current ranges and highlights HC ranges in bold.

UVS256 Current Ranges

		Current Ranges and Voltage Limits		
Channel Type	Leader Channel	-2, 2.5 V	-2, 7.2 V	-2, 18 V
DCVS	1, 2, 3..., 256	800mA (HC) 700mA (HC) 200mA 20mA 2mA 200A 20A 4A	200mA 20mA 2mA 200A 20A 4A	20mA 2mA 200A 20A 4A
DCVSMerged2	1, 3..., 255	1400mA (HC) 700mA (HC) 400mA 40mA 200mA 20mA 2mA 200A 20A 4A	400mA 40mA 200mA 20mA 2mA 200A 20A 4A	40mA 20mA 2mA 200A 20A 4A
DCVSMerged4	1, 5, 9..., 253	2800mA (HC) 700mA (HC) 200mA 20mA 2mA 200A 20A 4A	200mA 20mA 2mA 200A 20A 4A	20mA 2mA 200A 20A 4A

DCVSMerged8	1, 9..., 249	5600mA (HC)	200mA	20mA
		700mA (HC)	20mA	2mA
		200mA	2mA	200A
		20mA	200A	20A
		2mA	20A	4A
		200A	4A	
		20A		
		4A		

For more information on merged supplies, see:

- Merging Channels
- UVS256 Merged Model and Sense Circuitry

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UltraVS256 DCVS Programming : DCVS DataTool Programming

DCVS DataTool Programming

This section describes how to program several DataTool worksheets to handle basic DCVS parameters. It includes the following topics:

- Programming the Pin Map
- Programming the Channel Map
- Programming Pin Levels
- DCVS Test Instance Programming

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UltraVS256 DCVS Programming : DCVS DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. Select the Power pin type for each device pin name connected to a DCVS instrument.

For more information, see Pin Map Sheet in the DataTool section.

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UltraVS256 DCVS Programming : DCVS DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the DCVS instrument, you must select one of the four channel types from the Tester Channel Type column menu for each pin or pin group. The following table lists the available DCVS instrument channel type options.

DCVS Instrument Channel Type Options

Option	Explanation
DCVS	Standard connection to a single DCVS instrument on one UVS256 board. See UVS256 Current Ranges.
DCVSMerged2	Indicates a DIB wiring connection that merges two DCVS instruments on one UVS256 board. See UVS256 Current Ranges.
DCVSMerged4	Indicates a DIB wiring connection that merges four DCVS instruments on one UVS256 board. See UVS256 Current Ranges.
DCVSMerged6	Not supported on UVS256.
DCVSMerged8	Indicates a DIB wiring connection that merges eight DCVS instruments on one UVS256 board. See UVS256 Current Ranges.

When assigning channels for a specific site in the Tester Channel Site column, use the following channel map signal name format:

<slot_number>.sense<leader_vs_number>

where:

<slot_number>	DIB slot where the UVS256 board is installed
<leader_vs_number>	DCVS leader channel number. Channel numbers start with 1.

For example, if a UVS256 board is in DIB slot 19 and the leader channel number is 1, your channel map signal name is 19.sense1.

For high current (HC) channels, add the channel specification with an “hc” suffix in the channel map site entry. For example, if a UVS256 board is in DIB slot 5 and the leader channel number is 2, the signal name for an HC channel would be 5.sense2hc.

☑ **Note:**

Do not mix UVS256 HC and SC channels in a pin group. IG-XL returns a validation error for this invalid programming setup.

For more information, see:

- Channel Map Sheet
- UVS256 Merging
- Merging Channels
- UVS256 Channel Selection Guidelines

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UltraVS256 DCVS Programming : DCVS DataTool Programming : Programming Pin Levels

Programming Pin Levels

A Pin Levels sheet defines values for input and output voltages and current loads to be applied to the DUT. Once you insert this sheet into a test, you can program the Power Supply Test Instance Editor, which uses values assigned in this sheet.

For the DCVS instrument, you must assign a pin levels parameter for each pin name or group. The following table shows the UVS256 DCVS instrument pin level parameter options.

UVS256 DCVS Instrument Pin Levels Parameter Options

Option	Explanation
VMain	Main voltage power supply. Used to supply voltage to the DUT through the main DAC. See Selecting VMain, VAlt, and IFoldLevel.
VAlt	Alternate voltage power supply. Used to supply voltage to the DUT through the alternate DAC. See Selecting VMain, VAlt, and IFoldLevel.
IFoldLevel	Fold current limit level. See Selecting VMain, VAlt, and IFoldLevel and Setting the Current Limit.
IOverloadLevel	Optional. Overload current limit level. Not supported on UVS256. IG-XL returns an error if programmed on UVS256.
Tdelay	Time delay. Used to define the delay time after the application of the supply voltage. By programming Tdelay, you can wait some amount of time after applying device power. Tdelay is not used during the power-down sequence. It is only used after powering up all power pins of the same sequence number.

If you place more than one DCVS pin in the same sequence group on a Pin Levels sheet, then IG-XL applies power to all of those pins at once and uses the maximum of all Tdelay values for these pins.

For information, see Pin Levels Sheet in the DataTool section.

Selecting VMain, VAlt, and IFoldLevel

By default, the pin level auto current range selection is enabled, and UVS256 selects the highest possible current range based on IFoldLevel, VMain, and VAlt. Depending on which current range is selected, the maximum allowable voltage for VMain or VAlt can be 2.5V, 7.2V, or 18V.

The following voltages are valid for each listed current range (Irange):

- Voltage = [-2 to 18]V: Irange 4A, 20A, 200A, 2mA, 20mA, 40mA
- Voltage = [-2 to 7.2]V: Irange 200mA, 400mA
- Voltage = [-2, 2.5]V: Irange 700mA, 800mA, 1000mA, 2000mA, 4000mA

If no current range is available that supports the desired IFoldLevel, VMain, and VAlt, UVS256 returns a validation error.

In some cases, automatic selection of current range may be undesirable. For example, when changing between merged and unmerged ranges, BandwidthSetting should also be changed if it is not set to safe mode (255). Therefore, make sure that CurrentRange changes are carefully sequenced. For more information, see Safely Modifying BandwidthSetting and CurrentRange.

To turn off the pin level auto current range selection, you can use the following hidden VBT syntax:

TheHdw.DCVS.ApplyLevelsAutoIrange = False

When auto current range selection is turned off, however, you are responsible for programming a valid current range in VBT that fulfills the IFoldLevel, VMain, and VAlt parameters before calling:

TheHdw.Digital.ApplyLevelsTiming LoadLevels:=True

If you do not program a valid current range, UVS256 returns a run-time error when TheHdw.Digital.ApplyLevelsTiming LoadLevels:=True is called.

For more information, see the table in UVS256 Merging.

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UltraVS256 DCVS Programming : DCVS DataTool Programming : DCVS Test Instance Programming

DCVS Test Instance Programming

You can use the instance editor to create and edit DCVSPowerSupply_T VBT test instances. A test instance defines a test that is applied to the DUT when the test program runs. You can create as many test instances as you want. The Test Instances sheet lists all test instances for your test program.

Before creating a DCVS test instance, you must declare a Power pin type in the pin map. You also need a Pin Levels sheet. Finally, you must connect the Power pin to a DCVS channel type in the channel map.

To create or edit a DCVSPowerSupply_T test instance in the instance editor, see Creating and Editing Test Instances. Once you create a test instance, program the DCVSPowerSupply_T test instance by filling in the parameter fields. For more information, see DCVS Power Supply VBT Test.

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UltraVS256 DCVS Programming : DCVS VBT Programming

DCVS VBT Programming

Several basic DCVS programming tasks require that you use VBT syntax.

- Setting the Current Limit
- Setting the Output Compensation Bandwidth
- Setting Alarms
- Setting the DCVS Current Meter
- Setting Up DCVS Capture

For more information, see DCVS in the VBT Syntax Reference.

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UltraVS256 DCVS Programming : DCVS VBT Programming : Setting the Current Limit

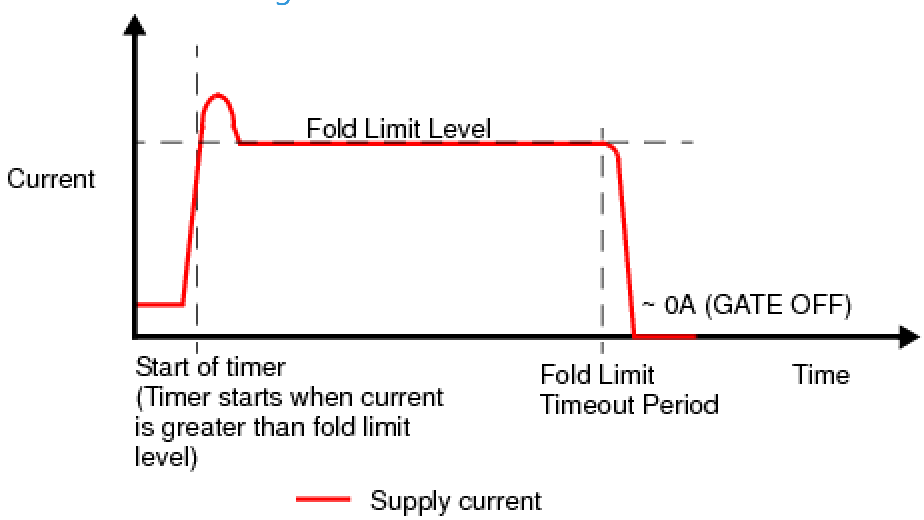
Setting the Current Limit

The UVS256 instrument has one single-level fold current limiting mode. Current limiting helps prevent damage to probe needles while high short-circuit current is flowing.

Note that the source and sink fold levels have different min and max limits than the current range for the instrument. For example: In 200mA range, the max source fold level is 200mA and the min source fold level is 25mA. The maximum sink fold level in this range is 75mA and minimum sink fold level is 25mA.

☑ **Note:**

The UVS256 does not support a dual-level overload current limit mode (overload limit enabled). The hardware limits the source or sink current to a programmed value until the user-defined fold time expires or the current drops below the limit level value. The following figure shows this single-level current limiting.



Single-Level Current Limiting

If the current is greater than the fold limit level, the fold limit timer starts, and the current is immediately limited to the fold limit level. (The timer has a resolution of 100s, and the maximum timeout is 2s.) The timer resets if the current drops below the fold limit level. If the timer is not reset and the fold limit timeout period expires, however, the hardware forces the DCVS output to gate off. Optionally, you can program the output to stay gated on for as long as possible after the fold limit timeout period has elapsed.

After determining the desired source and sink current limit types, you can program the fold limit levels and timeout values to activate the single-level mode. The following table lists UVS256 source and sink fold current limits.

UVS256 Fold Limits

	Source Fold Limits		Sink Fold Limits	
Current Range	Minimum	Maximum	Minimum	Maximum
5600mA	800mA	5600mA	800mA	800mA
2800mA	400mA	2800mA	400mA	400mA
1400mA	200mA	1400mA	200mA	200mA
800mA	100mA	800mA	100mA	100mA
700mA	100mA	700mA	100mA	100mA
400mA	50mA	400mA	50mA	150mA
200mA	25mA	200mA	25mA	75mA
40mA	5mA	40mA	5mA	40mA
20mA	2.5mA	20mA	2.5mA	20mA
2mA	0.25mA	2mA	0.25mA	2mA
200A	0.025mA	0.2mA	0.025mA	0.2mA
20A	0.0025mA	0.02mA	0.0025mA	0.02mA
4A	0.0005mA	0.004mA	0.0005mA	0.004mA

For more information, see Selecting VMain, VAlt, and IFoldLevel.

UVS256 Current Limit Settings

The default current limit settings when IG-XL is started are as follows:

Source Fold Limit Level	20mA
Source Fold Limit Timeout	1s

Source Fold Limit Behavior	Do not Gate Off
Sink Fold Limit Level	20mA
Sink Fold Limit Timeout	1s
Sink Fold Limit Behavior	Do not Gate Off

Note that the UVS256 shares one timer between source and sink fold limit timeout. This means that programming one timeout (source or sink) will overwrite the other timeout.

When the Pin Levels sheet levels settings are applied (for example, when using TheHdw.PinLevels.ApplyPower), the IFoldLevel parameter is used to set the source fold limit level. Timeout.Value and Behavior properties remain unchanged, and all sink parameters are provided with default values. You can change them using VBT.

To change the source current levels, you can use VBT syntax as shown in this example:

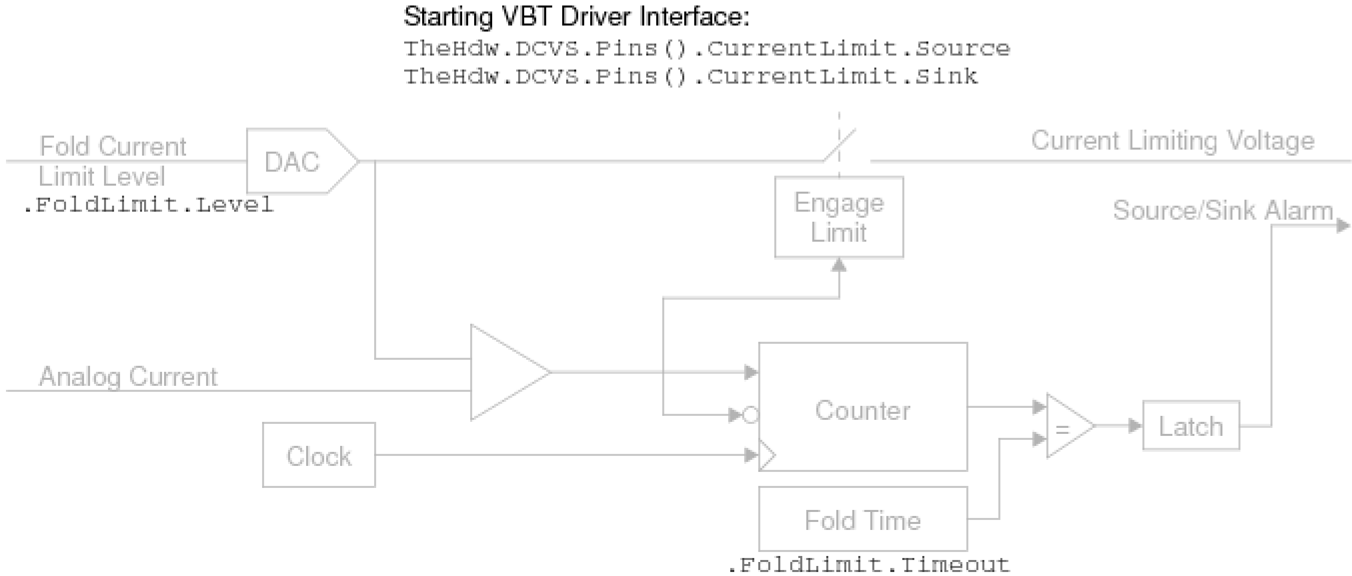
```
With TheHdw.DCVS.Pins(PinList)
  .Gate = False ' You must gate off to change fold limit timeout value
  .CurrentRange = 0.200
  .CurrentLimit.Source.FoldLimit.Level.Value = _
     (0.65 * .CurrentLimit.Source.FoldLimit.Level.Max)
  .CurrentLimit.Source.FoldLimit.Timeout.Value = 0.010
  .CurrentLimit.Source.FoldLimit.Behavior = tlDCVSCurrentLimitBehaviorGateOff
End With
```

To change the sink current levels, you can use VBT syntax as shown in this example:

```
With TheHdw.DCVS.Pins(PinList)
  .Gate = False ' You must gate off to change fold limit timeout value
  .CurrentRange = 0.200
  .CurrentLimit.Sink.FoldLimit.Level.Value = 0.075
  .CurrentLimit.Sink.FoldLimit.Timeout.Value = 0.010
  .CurrentLimit.Sink.FoldLimit.Behavior = tlDCVSCurrentLimitBehaviorGateOff
End With
```

You can use an OnProgramStarted executive interpose function to set up your own start-of-program current limiting “defaults” for an entire test program run.

The following figure shows how the VBT statements program the hardware components.



DCVS Instrument: Programming Current Limiting

Best Programming Practice When Choosing Fold Level and Current Range

UVS256 channels are arranged in four-channel groups (ch1 - ch4, ch5 - ch8, and so on), and you can designate any channels in a group as HC. When programming source fold levels, however, the fold levels in any four-channel group cannot exceed 2.8A.

When you specify IFoldLevel on a Pin Levels sheet and if pin level auto current range selection is enabled, UVS256 selects the highest possible current range that fulfills the values for IFoldLevel, VMain, and VAlt. See Selecting VMain, VAlt, and IFoldLevel.

Although IG-XL attempts to pick an appropriate Irange, the range chosen may not always provide the best performance from the instrument. In such cases, you can use VBT syntax to pick the most appropriate range yourself, overriding the range derived from the Pin Levels sheet. The following is an example:

TheHdw.DCVS.Pins(A).CurrentRange.Value = 800mA

TheHdw.DCVS.Pins(A).CurrentLimit.Source.FoldLimit.Level.Value = 750mA

Here, both the current range and the fold current limit level are set using VBT statements, overriding the Pin Levels sheet. If you then include code to reduce the current range, IG-XL automatically reduces the fold source level appropriately:

TheHdw.DCVS.Pins(A).CurrentRange.Value = 20mA ' SW reduces fold level to 20mA

x = Pins(A).CurrentLimit.Source.FoldLimit.Level.Value ' x is 20mA

However, suppose you change the current range again back to the higher 800mA value:

Pins(A).CurrentRange.Value = 800mA

x = Pins(A).CurrentLimit.Source.FoldLimit.Level.Value

What will x be now? In this case, IG-XL does not reset the fold level to 750mA, but to 100mA, which is the minimum source fold level for the 800mA current range. This is a level that you may not expect and that may not be optimal for your device.

Therefore, if you program the UVS256 force current value using VBT, consider modifying the fold level value at the same time.

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UltraVS256 DCVS Programming : DCVS VBT Programming : Setting the Output Compensation Bandwidth

Setting the Output Compensation Bandwidth

The bandwidth parameter allows the UVS256 instrument to provide optimal performance with various output current range modes and DIB capacitance. These settings allow you to select the desired performance with your DIB and bypass capacitors.

The DPS FPGA in the UVS256 has selectable resistance and capacitance in its control loop which affects control loop frequency response. Depending on the DIB capacitance and the test requirements, you can tune these values for optimal instrument performance. The BandwidthSetting VBT parameter allows you to specify a code to the driver for controlling this setup.

The BandwidthSetting code is a mathematical combination of a code corresponding to the bypass capacitance on the DIB, and the ESR of the capacitors. Selecting the BandwidthSetting Code outlines how to compute this code.

Choose capacitance values to trade off droop (larger capacitance can respond faster to changing current demands) against current noise performance. (Choose smaller capacitance to reduce current noise.) The programmable BandwidthSetting helps you to optimize the selected capacitance. For guidelines on selecting capacitance, see Bypass Capacitance in the UVS256 section of DIB Design Guidelines.

Note that changes to the output current range affect the performance of a given bandwidth. The DCVS meets or exceeds its dynamic performance specifications if the power supply’s sense lines are at the DIB capacitance and if you meet the minimum design specifications for the DIB and its capacitance. To ensure stability, connect the sense line to the force line as close as possible to the bypass capacitor, and connect the DGS to “device ground” as close as possible to the capacitor. Also connect the bypass capacitors as close as possible to the DUT. For more information, see UltraVS256 DIB Design Guidelines.

Selecting the BandwidthSetting Code

The BandwidthSetting code specifies capacitive load on the UVS256 pin. This code depends on a combination of:

- DIB bypass capacitance
- DIB bypass equivalent series resistance (ESR)

The process for choosing a code is:

1. Determine if the range you are using is an unmerged or a merged range.
2. Use the DIB bypass capacitance to determine the C code value.
3. Use the DIB bypass equivalent series resistance (ESR) to determine the ESR code value.
4. Calculate the BandwidthSetting and determine if that value is valid for the instrument.

The following table shows the available current ranges. If the programmed current range is an unmerged range, treat it as such for purposes of BandwidthSetting programming, even if it has access to merged ranges. Note that this means that if a range is changed from an unmerged to a merged range, BandwidthSetting should be recalculated.

UVS256 Current Ranges

Current Range	HC?	Mode
4 A	No	Unmerged
20 A	No	Unmerged
200 A	No	Unmerged
2 mA	No	Unmerged
20 mA	No	Unmerged
40 mA	No	DCVSMerged2
200 mA	No	Unmerged
400 mA	No	DCVSMerged2

700 mA	Yes	Unmerged
800 mA	Yes	Unmerged
1.4 A	Yes	DCVSMerged2
2.8 A	Yes	DCVSMerged4
5.6 A	Yes	DCVSMerged8

Use this table to determine whether you are using an unmerged or a merged range. Note that changing from some ranges to other ranges changes the mode you are in. In these cases, you must reprogram BandwidthSetting. (See Safely Modifying BandwidthSetting and CurrentRange.)
Next, use the following table to find the "C code value." Note that any capacitance in bold results in an out-of-range BandwidthSetting. In this case, the instrument will use Safe Mode (255).

C Code Values for BandwidthSetting

C Code Value	Capacitance (F)							
	Unmerged		DCVSMerged2		DCVSMerged4		DCVSMerged8	
	Min	Max	Min	Max	Min	Max	Min	Max
0	0	0.05	0	0.1	0	0.2	0	0.4
1	0.05	0.083	0.1	0.166	0.2	0.332	0.4	0.664
2	0.083	0.138	0.166	0.276	0.332	0.552	0.664	1.104
3	0.138	0.229	0.276	0.458	0.552	0.916	1.104	1.832
4	0.229	0.38	0.458	0.76	0.916	1.52	1.832	3.04
5	0.38	0.63	0.76	1.26	1.52	2.52	3.04	5.04
6	0.63	1.1	1.26	2.2	2.52	4.4	5.04	8.8
7	1.1	1.7	2.2	3.4	4.4	6.8	8.8	13.6
8	1.7	2.9	3.4	5.8	6.8	11.6	13.6	23.2
9	2.9	4.8	5.8	9.6	11.6	19.2	23.2	38.4
10	4.8	7.9	9.6	15.8	19.2	31.6	38.4	63.2
11	7.9	13	15.8	26	31.6	52	63.2	104
12	13	22	26	44	52	88	104	176
13	22	36	44	72	88	144	176	288

14	36	60	72	120	144	240	288	480
15	60	160	120	320	240	640	480	1280

Next, use the following table to find the ESR code value. Note that any ESR in bold results in an out-of-range BandwidthSetting. In this case, the instrument will use Safe Mode (255).

ESR Code Values for BandwidthSetting

ESR Code Value	ESR (mohms)							
	Unmerged		DCVSMerged2		DCVSMerged4		DCVSMerged8	
	Min	Max	Min	Max	Min	Max	Min	Max
0	0	1	0	0.5	0	0.25	0	0.125
1	1	1.8	0.5	0.9	0.25	0.45	0.125	0.225
2	1.8	3.4	0.9	1.7	0.45	0.85	0.225	0.425
3	3.4	6.3	1.7	3.15	0.85	1.575	0.425	0.7875
4	6.3	12	3.15	6	1.575	3	0.7875	1.5
5	12	21	6	10.5	3	5.25	1.5	2.625
6	21	40	10.5	20	5.25	10	2.625	5
7	40	74	20	37	10	18.5	5	9.25
8	74	140	37	70	18.5	35	9.25	17.5
9	140	250	70	125	35	62.5	17.5	31.25
10	250	460	125	230	62.5	115	31.25	57.5
11	460	860	230	430	115	215	57.5	107.5
12	860	1500	430	750	215	375	107.5	187.5
13	1500	2900	750	1450	375	725	187.5	362.5
14	2900	6400	1450	3200	725	1600	362.5	800
15	6400	10000	3200	5000	1600	2500	800	1250

Finally, you compute the BandwidthSetting to program using the following equation:
BandwidthSetting_proposed = 16 * (C code value) + (ESR code value)
The computed BandwidthSetting_proposed must fall within a safe range, as listed below. These regions correspond to safe regions in the C code value and ESR code value tables above. If

BandwidthSetting_proposed is not in a safe range, program 255 (safe mode).

Valid BandwidthSetting Values

Range	Unmerged	DCVSMerged2	DCVSMerged4	DCVSMerged8
0-175	OK	not allowed -use safe mode		
176-181	OK	OK	OK	OK
182-191	OK	not allowed - use safe mode		
192-197	OK	OK	OK	OK
198-207	OK	not allowed - use safe mode		
208-213	OK	OK	OK	OK
214-223	OK	not allowed - use safe mode		
224-229	OK	OK	OK	OK
230-239	OK	not allowed - use safe mode		
240-245	OK	OK	OK	OK
246-254	OK	not allowed - use safe mode		
255 (Safe Mode)	OK	OK	OK	OK

Safely Modifying BandwidthSetting and CurrentRange

Use the following syntax to specify the selection of compensation bandwidth:

TheHdw.DCVS.Pins.BandwidthSetting.Value = 0 - 255

A setting of 255 is the instrument's "safe" mode.

Control loop frequency response is set by the combination of the programmed BandwidthSetting and CurrentRange. Because you can program CurrentRange and BandwidthSetting independently, take care in sequencing these parameters.

For device pins that have access only to unmerged (single channel) resources, BandwidthSetting should only need to be applied once in a test program (for example in OnProgramStarted). Use this sequence:

```
With TheHdw.DCVS.Pin(pin)
   .BandwidthSetting.Value = Bandwidth
   .CurrentRange.Value = iRange
End With
```

For device pins that have access to merged channel resources, take care to identify when the programmed current range is modified from using an unmerged range to using a merged range, and vise-versa. These changes generally require a careful change to the BandwidthSetting. Note that pins may oscillate if you apply an incorrect BandwidthSetting. (Typically, this is as a result of improper sequencing.)

Use the following sequence when setting CurrentRange and BandwidthSetting for the first time:

```
' initial setting
With TheHdw.DCVS.Pins(pin)
    .BandwidthSettings.Value = BW_init
    .CurrentRange.Value = IRange
End With
```

Use the following sequence when changing from an unmerged to a merged range or when changing from a merged to an unmerged range:

```
With TheHdw.DCVS.Pin(pin)
    .BandwidthSetting.Value = 255 ' safe mode
    .CurrentRange.Value = IRange
    .BandwidthSetting.Value = New_Bandwidth
End With
```

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UltraVS256 DCVS Programming : DCVS VBT Programming : Setting Alarms

Setting Alarms

An alarm indicates a hardware condition that may invalidate a measurement. These conditions can be caused by a bad device, an incorrectly programmed instrument, or a test system failure. By default, a test is forced to fail if an alarm occurs. You can set the action to be taken for each DCVS pin when an alarm occurs. The test can be forced to bin, or the alarm can be ignored. For more information, see Alarms Subsystem.

To set up alarms for a test program, use the following VBT syntax:

TheHdw.DCVS.Pins(Pinlist).Alarm(AlarmType) = AlarmBehavior

The following table shows the DCVS instrument alarm types for the UVS256.

DCVS Alarm Types for UVS256

Alarm Type	Description
tlDCVSAlarmAll	Activates all DCVS alarms.
tlDCVSAlarmCaptureNotTriggered	Indicates that the requested capture could not be started.
tlDCVSAlarmMemoryFull	Indicates that the capture memory is full.
tlDCVSAlarmOpenKelvinDIB	Monitors the voltage drop between the GND and DGS points. A voltage drop greater than the defined threshold between Low Sense (DGS) and DUT ground (GND) triggers this alarm.

tlDCVSAlarmOpenKelvinDUT	Monitors the voltage drop between the High Force and High Sense points. A voltage drop greater than the defined threshold between High Force and High Sense triggers this alarm.
tlDCVSAlarmOverrange	Indicates that a meter voltage or current sample exceeds the selected meter range.
tlDCVSAlarmFoldCurrentLimitTimeout	Indicates that the sink or source fold current limit has timed out.
tlDCVSAlarmSourceFoldCurrentLimitTimeout	Indicates that the source fold current limit has timed out.

This next table shows the DCVS instrument alarm behaviors. For more information, see Programming Instrument Alarm Behavior.

DCVS Alarm Behaviors

Alarm Behavior	Description
tlAlarmContinue	Report the alarm but do not change DUT status.
tlAlarmDefault	Restore behavior to default.
tlAlarmForceBin	Bin the current part.
tlAlarmForceFail	Current part fails.
tlAlarmOff	Disable alarms if possible.

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UltraVS256 DCVS Programming : DCVS VBT Programming : Setting the DCVS Current Meter

Setting the DCVS Current Meter

The UVS256 instruments have one set of current ranges for both forcing and metering current. Therefore, the force current range you set determines the range used to meter current.

You can use the meter filter for accurate hardware averaging, which reduces test time compared to the standard software averaging. To do this, the filter automatically performs multiple measurements (according to the integration time) and builds the average in the hardware. As a result, the current meter can calculate the average without requiring that you manually make multiple measurements.

Note the following:

- The source and current meters for the UVS256 are shared.
- Overcurrent meter range alarms will occur when the averaged value is exceeded, not when any single value is exceeded.

For more information, see Metering and Capture Subsystem.

Setting Current Ranges

UVS256 has a single current range setting per channel that applies to sink and source current fold limit levels and to full-scale metering current capability. The programming model used for setting current ranges for the instrument is to ensure compatibility with other DCVS instruments, such as HDVS and HexVS.

To change the common force and meter current range, use the properties:

TheHdw.DCVS.Pins.CurrentRange.Value

TheHdw.DCVS.Pins.PSets.CurrentRange.Value

To get a list of programmable current ranges, use the properties:

TheHdw.DCVS.Pins.CurrentRange.List

TheHdw.DCVS.Pins.PSets.CurrentRange.List

To get or set the meter current range, use the property:

TheHdw.DCVS.Pins(<PinList>).Meter.CurrentRange.Value

Note that you can only program the meter current range to the presently set current range value. Setting the meter current range to another value results in a run-time error.

Strobing and Reading the Meter

To strobe the meter, use the method:

TheHdw.DCVS.Pins.Meter.Strobe(SampleSize, SampleRate)

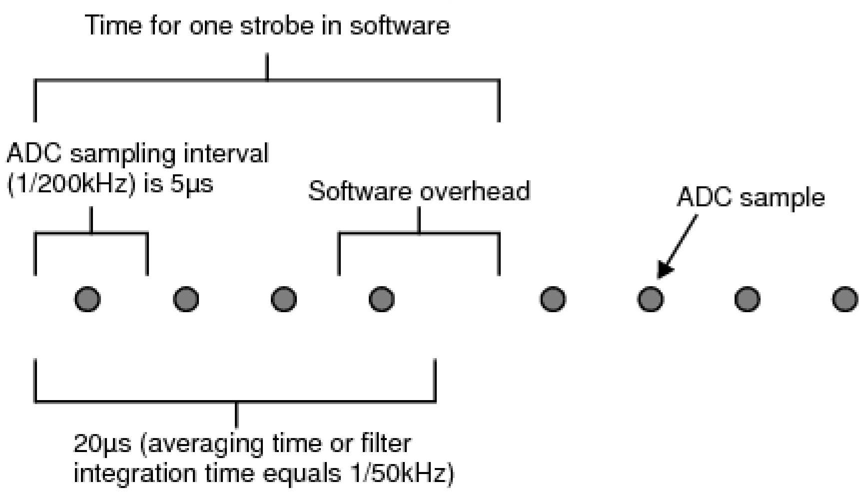
where SampleSize specifies the number of strobes for all pins in the pin list. Each meter strobe is an average of multiple ADC samples. The meter filter value indirectly dictates the number of ADC samples to be used for averaging. The higher the filter value, the fewer ADC samples are averaged for each strobe. See DCVS.VBT.Meter.Filter.Value for a list of filter values supported by UVS256 and their corresponding number of ADC samples for averaging.

The parameter SampleRate is fixed at 200kHz. This parameter is available for programming for compatibility only and is ignored by the software.

Strobe data is invalid if the measurement range or the meter filter value is changed during meter integration time. Changing the meter mode does not affect measurements in progress but affects subsequent measurements.

The time needed for N strobes is pin-sites independent and is roughly equal to N * T_filter, where T_filter is the meter averaging time (or filter integration time) and is given by 1/FilterValue.

Take the case where N = 2 and Filter.Value = 50kHz. Given the fixed ADC sample rate of 200kHz, this means an ADC sample is collected every 5s and that the hardware will average four ADC samples for every strobe (200kHz /50kHz = 4 ADC samples). The averaging time (or filter integration time) equals 20s (1/50kHz), and this is the time needed to complete the averaging of four ADC samples. The following diagram illustrates the example.



Strobe Example

The Meter.Read method reads consecutively N strobed values from the current meter read-pointer location in FIFO order. If more samples are read than the available strobed values in meter memory, the overrun samples contain whatever values stay in memory. No error or alarm is raised in this case.

For the Meter.Read method, SampleRate is fixed at 200kHz. This parameter is available for programming for compatibility only and is ignored by the software.

A current strobed value passes if it is within the lower and upper meter current test limits. Otherwise, it fails. The lower and upper current test limits are given by the properties Meter.TestLimits.ILo.Value and Meter.TestLimits.IHi.Value.

A voltage strobed value passes if it is within the lower and upper meter voltage test limits. Otherwise it fails. The lower and upper voltage test limits are given by the properties Meter.TestLimits.VLo.Value and Meter.TestLimits.VHi.Value.

Setting the Meter Filter

To get or set the meter filter value, use the property:

TheHdw.DCVS.Pins.Meter.Filter.Value

This statement programs the meter filter value to all pins in the pin list at all selected active sites. If no meter filter value matches the specified filter, then the next higher filter value is selected.

UVS256 uses simple discrete-time filtering (N-point running averager) to filter noise. The instrument supports averaging 0 to 12 powers of two samples collected at a fixed 200kHz ADC sample rate. Instead of specifying the number of samples N to be averaged, you specify a filter frequency. The filter frequency is the ADC sample rate divided by the number of samples being averaged. The relationship between the filter frequency, number of samples N and averaging time (or filter integration time) is:

FilterFrequency = ADCSampleRate / N

= 1 / (N / ADCSampleRate)

= 1 / (N * T), where T is ADC sampling interval and is given by 1 / ADCSampleRate

= 1 / AveragingTime

The table below shows the number of samples and the corresponding filter frequency supported.

UVS256 Filter Frequencies

Number of Samples	Filter Frequency
1	200 kHz
2	100 kHz
4	50 kHz
8	25 kHz

16	12.5 kHz
32	6.25 kHz
64	3.125 kHz
128	1.5625 kHz
256	781 Hz
512	391 Hz
1024	195 Hz
2048	98 Hz
4096	49 Hz

You can enable or disable the meter filter using the Pins.Meter.Filter.Bypass property. This statement programs the meter filter bypass to all pins in the pin list at all selected active sites. If the statement is set to False, then meter filter is not bypassed, and every meter strobe is a result from averaging of multiple meter measurements taken within the meter filter integration time. Meter filter integration time is given by 1/Meter.Filter.Value. If meter filter is bypassed (True), then every meter strobe is a single meter measurement.

For more information, see UltraVS256 Specifications.

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UltraVS256 DCVS Programming : DCVS VBT Programming : Setting Up DCVS Capture

Setting Up DCVS Capture

To get and set the capture sample rate in Hz, use the property:

TheHdw.DCVS.Pins.Capture.SampleRate.Value

To program the capture sample rate to all pins in the pin list at all selected active sites, use the statement:

TheHdw.DCVS.Pins<PinList>.Capture.SampleRate.Value

If no capture sample rate matches the programmed sample rate, then the next higher sample rate is selected.

Valid sample rate settings are programmed in Hertz (maximum = 200000, minimum = 48). These values are based on the capture ADC's fixed 200kHz sample rate. IG-XL always programs the sample rate as a sub-multiple of 200kHz. For example, if the desired sample rate is 43kHz, then the actual sample rate is 200kHz/43kHz, or 4.6511kHz. IG-XL truncates this value to 4KHz, so the programmed sample rate is 200kHz/4kHz, or 50kHz, which is the next highest sample rate.

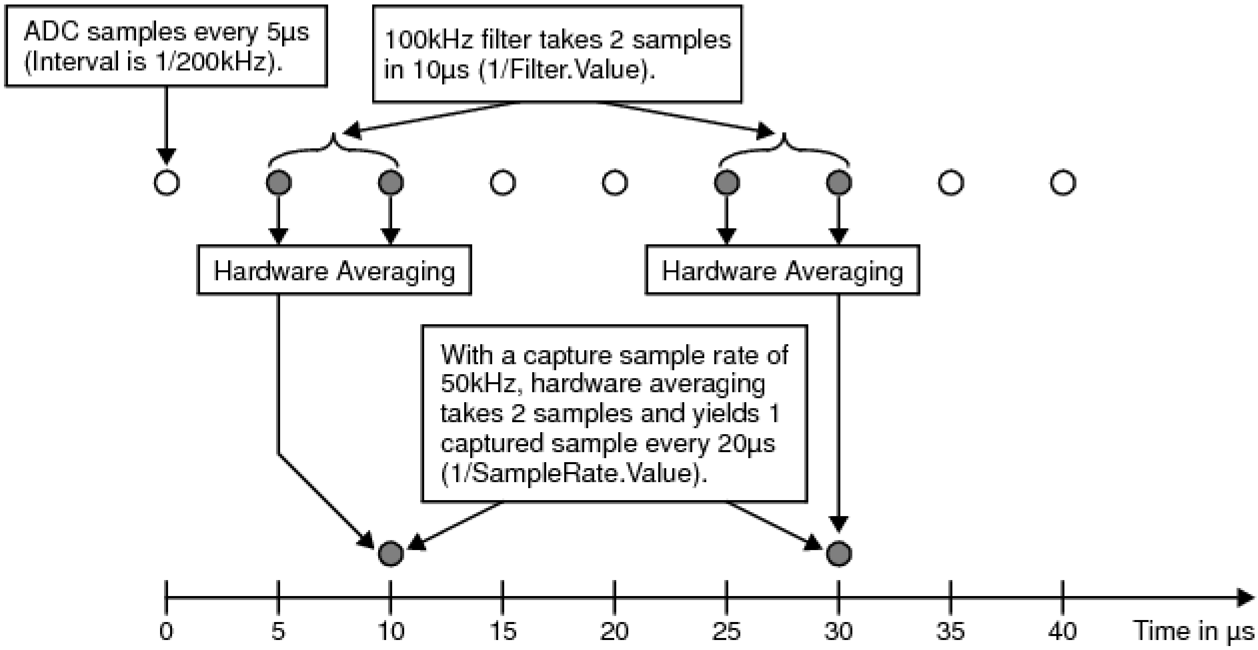
You can also determine the sample rate using the UVS256's hardware averager, which provides ranges from 1 to 4096, where values are powers of two (1, 2, 4, 8, and so on). IG-XL uses these filter setting values to determine the number of averages to use for each returned capture value. The table below shows these values:

UVS256 Sample Rate by Hardware Averaging

VBT Programmed Filter Rate	Averages per Sample
200kHz	200kHz/200kHz = 1
100kHz	200kHz/100kHz = 2
50kHz	200kHz/50kHz = 4

25kHz	200kHz/25kHz = 8
12.5kHz	200kHz/12.5kHz = 16
6.25kHz	200kHz/6.25kHz = 32
3.125kHz	200kHz/3.125kHz = 64
1.5625kHz	200kHz/1.5625kHz = 128
781.25Hz	200kHz/781.25Hz = 256
390.625Hz	200kHz/390.625Hz = 512
195.3125Hz	200kHz/195.3125Hz = 1024
97.65625Hz	200kHz/97.65625Hz = 2048
48.828125Hz	200kHz/48.828125Hz = 4096

To get or set the capture sample size to all pins in the pin list at all selected active sites, use the property:
TheHdw.DCVS.Pins.Capture.SampleSize.Value
To trigger a capture signal in VBT, use the method:
TheHdw.DCVS.Pins.Capture.Signals.Trigger
This statement triggers UVS256 to capture N signal samples at the given capture sample rate for all pins in the pin list at all selected active sites. Each signal sample is an average result from one or multiple ADC samples. The meter filter frequency determines the number of ADC samples to be used for averaging. See DCVS.VBT.Meter.Filter.Value for the list of filter values supported by UVS256 and their corresponding number of ADC samples for averaging.
The following diagram shows a case where the ADC sampling interval is 5s, the sample rate is 50kHz (20s), and the filter frequency is 100kHz (10s). This means that 2 ADC samples will be averaged.



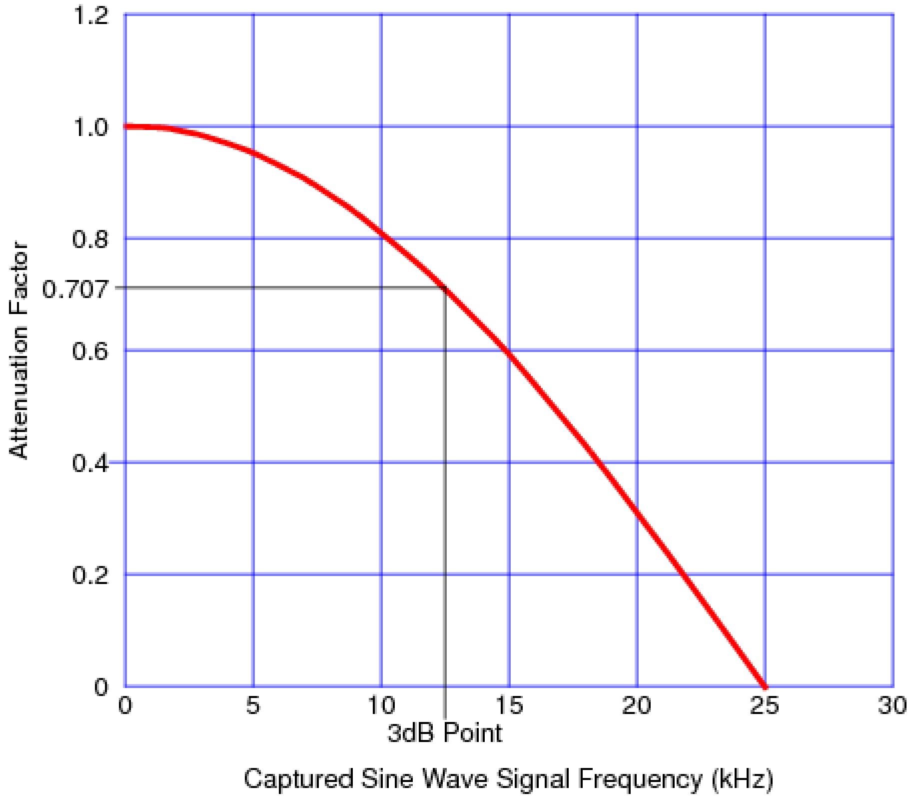
Capture Example

Make sure to set the filter frequency to a value greater than the capture sample rate. Also make sure that the capture signal's settings are loaded before a trigger from VBT or from a pattern. Changing the capture sample rate, capture sample size, or meter mode during a capture does not affect the capture in progress. Changing the current range, voltage range, or meter current range during capturing current results in invalid data. Only a single capture per pin at a time is supported. Any trigger that happens on the same pin during a capture is ignored and results in an alarm tlDCVSAlarmCaptureNotTriggered condition on that pin. In this case, the DspWave will have no data.

A meter strobe on the same pin during a capture is ignored and results in an alarm tlDCVSAlarmCaptureNotTriggered condition on that pin.

Capture Path Bandwidth Limitations

When using this capture setup syntax, filtering actually uses hardware averaging to remove noise by averaging multiple 200kHz samples. This moving average filter is good for removing random noise when monitoring the supply's current or voltage. In this case, the amount of noise reduction is equal to the square root of the number of samples averaged. While this type of filter is good for noise reduction, it can adversely affect edges because of the resulting attenuation of the measured capture voltage as the captured signal frequency increases. The graph below shows an example of this attenuation when measuring sine wave amplitudes using settings from the capture example above, which samples at 50kHz, filters at 100kHz, and averages two ADC samples.



UVS256 Attenuation Factor vs. Captured Sine Wave Signal Frequency

Based on the UVS256 capture path's attenuation factor, you can expect a smaller measured voltage as the frequency increases. For instance, if the input DC voltage is 3V with an attenuation factor of 1, then the resulting measured voltage at capture is 3V. But if this same 3V is measured at a frequency of 15kHz, then the resulting capture voltage is only 1.8V, since the value is reduced by an attenuation factor or 0.6.

This example shows that the UVS256 capture path bandwidth is limited, with the 3dB point at around 12kHz. So, if your device output has higher frequency components, then the capture path's inherent bandwidth limitation may filter them out, make the device output look significantly better than it is, and result in the possibility of passing bad devices, depending on the application's use case. Consequently, you should not rely on this instrument to provide usable results on total harmonic distortion (THD) or other sine wave related measurements. To avoid this attenuation effect on the UVS256's capture path, you would have to add an appropriate low pass filter on the DIB and set the capture filter to 200kHz. Because of these constraints, do not use the UVS256 as a general purpose digitizer. If you need this functionality, then use a waveform digitizer such as the UltraPAC80.

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UltraVS256 DCVS Programming : UVS256 Reset Values

UVS256 Reset Values

Resetting UVS256 connections disconnects the UVS256 from the DUT. Resetting the settings clears the UVS256 memory and returns the UVS256 to the default settings.

The table shows the reset state of UVS256 parameters after a reset occurs.

UltraVS265 Reset Values

Parameter	Reset Value
All Channels Gate	Off (force 0V)
All Channels HiZ	Not HiZ
Channel Disconnect	Disconnected
Current Range	20mA
Channel Output	Main
Main Value	0V
Alt Value	0V
Filter	Off
Bandwidth	255 (this corresponds to “safe” mode)
Meter Mode	Measure Current
Signals Sample Rate	200 kHz
Source Fold Level	20mA
Sink Fold Level	20mA

Merge Mode	No Merge
Fold Timer	1s
Fold Behavior	Do not Gate Off
Alarm States	Alarms Cleared
Alarm Status	Alarms Enabled
Measure Memory	Empty (don't expend time clearing this memory)
Signals Memory	Empty (don't expend time clearing this memory)
Cal connect	Disconnected

Instrument Disconnect State

If you disconnect the UVS256 instrument, it goes into a disconnect state with the following values:

- Current range: 20mA
- Source Fold Level: 20mA
- Sink Fold Level: 20mA
- Gate: Off
- Mode: Force V
- Bandwidth: 255 (safe mode for compensation)

In this state, the instrument will not respond to any changes that you make using VBT or the DCVS Debug Display. Once the instrument is reconnected, any changes that you made using VBT or the debug display will be applied in a specific order. If no changes were made since disconnect, the instrument will return to the state it was in before it was disconnected.

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UltraVS256 DCVS Programming : DCVS Pattern Control

DCVS Pattern Control

This section describes how to create DCVS test patterns by writing your own test patterns in an ASCII text file using the digital pattern format and DCVS microcodes. You can also use parameter sets (PSets) to program a DCVS instrument's state for testing.

Use the PatternTool to display and debug DCVS test patterns. To bring up PatternTool:

- Click Pattern Tool in the One-Click Launch group on the IG-XL Tools tab.
- From the Windows Start menu, select Programs>Teradyne IG-XL <version>>Tools>Pattern Tool.

This section includes the following topics:

- UVS256 DCVS Pattern Microcodes
- DCVS Pattern File Example
- PSets

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UltraVS256 DCVS Programming : DCVS Pattern Control : UVS256 DCVS Pattern Microcodes

UVS256 DCVS Pattern Microcodes

To create simple or complex test patterns, you can write your own ASCII text file using the digital pattern format and DCVS microcodes. DCVS microcodes can be used to control DCVS settings during pattern execution. This section describes pattern microcodes that are used only in DCVS patterns.

In DCVS patterns:

- All microcodes in a pattern vector are executed when the pattern vector is run. If a microcode in a pattern does not have all of its fields programmed, the unprogrammed fields are programmed to the default values when the pattern vector is run.
- You cannot have more than one DCVS microcode for each pin.
- Microcodes are not case sensitive.

Use the following DCVS microcodes in test pattern files:

UltraVS256 DCVS Pattern Microcodes

Microcode	Description
Nop	No operation (does nothing).
Pset <PSet Name>	Applies all parameter settings for the specified PSet to the hardware.
Gate_On	Gates the DCVS on.
Gate_Off	Gates the DCVS off.

Strobe	Strobes the DCVS meter once for that pin or channel. The reading is stored in measurement RAM. To retrieve use the call TheHdw.DCVS.Pins("pinname").Meter.Read(tlNoStrobe, 1). By specifying no strobe, you read the measurement stored in memory without taking a new measurement.
Enable_Alarm	Enables reporting of all the DCVS alarms.
Disable_Alarm	Disables reporting of all the DCVS alarms while the PatGen is running.
VS_Main_Sel	Selects the main voltage DAC. This value is the main voltage value you programmed before the pattern started, unless a PSet has been applied, in which case that voltage value is used.
VS_Alt_Sel	Selects the alternate voltage DAC. This value is the alternate voltage value you programmed before the pattern started, unless a PSet has been applied, in which case that voltage value is used.
Trig <Signal Name>	Triggers a capture with the specified capture tag and the current hardware settings. Since the LoadSettings method cannot be called during pattern execution, make sure to set up the hardware before the pattern starts. You can set up capture hardware settings during a pattern burst using PSets.
Filter_On	Turns the meter filter on.
Filter_Off	Turns the meter filter off.

For more information, see Microcode (Opcodes and Control Bits).

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UltraVS256 DCVS Programming : DCVS Pattern Control : DCVS Pattern File Example

DCVS Pattern File Example

The following DCVS pattern file shows the required pattern file statements. Note the following guidelines.

To make porting patterns between digital instruments easier, do not include a digital_inst or opcode_mode compiler options in your pattern file. Instead, include these options in the compile scripts or batch files. By following this programming practice, your pattern files are not locked into a particular instrument or timing mode and can be ported easily.

UltraFLEX UVS256 pattern microcodes can be executed from both vector memory (VM) and subroutine memory (SRM).

For an example of a pattern file used in Vbump testing, see Vbump Testing.

Settle Time Repeat Guidelines

The number of repeat cycles used in a settle time repeat depends on the following:

- The timeset period defined in the active time sheet
- The microcode or PSet settle time
- The instrument's analog settling time (if applicable)
- The minimum execution time required between microcodes or PSets

The goal is to have enough repeats to ensure that no microcode is executed before the previous one is completed and/or the new instrument's analog parameters have time to settle (for example, output voltage level). The microcode and PSet execution times are specified in the instrument board specification document.

```
import tset ts1;

instruments = {
    vdd : DCVS;
    vio : DCVS;
}
srm_vector ($tset, dig1:S)
{
    > ts1   1;
    > ts1   1;
    // ___ GATE ON ___
    (vdd : DCVS = Gate_On)
    > ts1   1;
    // settle time repeat. See guidelines above.
    // Note: the repeat count must be adjusted according the period of ts1.
    // So, if the period of ts1 is 100ns, then a repeat count of 5000 results
    // in a settling time of 100ns * 5000 = 0.5ms.
    repeat 5000
    > ts1   1;
    > ts1   1;
    > ts1   1;
    > ts1   1;
    // ___ GATE OFF ___
    (vio : DCVS = Gate_Off)
    > ts1   1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1   1;
    > ts1   1;
    > ts1   1;
    > ts1   1;
    // __ TRIGGER CAPTURE: my_capture_signal_label defined in job thru VBT __
    (vdd : DCVS = Trig my_capture_signal_label)
    > ts1   1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1   1;
    > ts1   1;
```

```
> ts1   1;
> ts1   1;
// ___ MULTIPLE PINS SAME MICROCODE ___
((vdd,vio) : DCVS = Gate_On)
> ts1   1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1   1;
> ts1   1;
> ts1   1;
> ts1   1;
// ___ MULTIPLE PINS DIFFERENT MICROCODES ___
(vdd : DCVS = VS_Alt_Sel)
(vio : DCVS VS_Main_Sel)
> ts1   1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1   1;
> ts1   1;
> ts1   1;
> ts1   1;
// ___ SINGLE STROBE ___
(vdd : DCVS = Strobe)
> ts_strobe   1;
> ts1   1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1   1;
> ts1   1;
// ___ PARAMETER SET (my_pset_1, my_pset_2 defined in job thru VBT ___
> ts1   1;
(vdd : DCVS = PSet my_pset_1)
> ts1   1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1   1;
> ts1   1;
> ts1   1;
```

```
    > ts1   1;
    (vdd : DCVS = PSet my_pset_2)
    > ts1   1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1   1;
    > ts1   1;
    > ts1   1;
    > ts1   1;
    > ts1   1;
}
```

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UltraVS256 DCVS Programming : DCVS Pattern Control : PSets

PSets

Parameter sets (PSets) allow you to program the state of an instrument for testing, such as the force voltage value and force voltage range. You set up its characteristics using its parameters. You define a PSet in VBT language and invoke it using PSet microcode.

Once defined, PSets can be applied through VBT or through patterns using the DCVS PSet microcode. Assuming you have defined a parameter set named, "my_pset", the following example shows how to apply the PSet from VBT:

```
TheHdw.DCVS.Pins("vdd").Psets("my_pset").Apply
```

The statement above is equivalent to:

```
TheHdw.DCVS.Pins("vdd").Psets.Item("my_pset").Apply
```

PSet parameters are initialized with their default values, and all parameters are updated when the PSet is applied.

The following example shows an ASCII test pattern with a DCVS microcode entry to apply this parameter set:

```
(vdd:DCVS = Pset my_pset)
   > tset0    H L 1 0 . . .;
```

PSets added to a high current (HC) pin are high current, and PSets added to a non-high current pin are low current.

The following UVS256 parameters can be set in a PSet:

UVS256 Parameters Available in PSets

PSet Parameter	Default Value
BandwidthSetting	Output compensation bandwidth. Default is 255.
CurrentRange	Output current range. Default is 20mA.

Meter Mode	Whether the DCVS is metering voltage or current. Default is voltage mode.
Capture SampleRate	Capturing sample rate. Default is 49.
Capture SampleSize	Capturing sample size. Default is 1.
Source Current FoldLimit	Source current fold limit. Default is 20mA.
Sink Current FoldLimit	Source current fold limit. Default is 20mA.
Main Voltage	Main output force voltage. Default is 0 V.
Alt Voltage	Alternate output force voltage. Default is 0 V.
Meter filter	Filter on the DCVS meter. Default is 49.
Output Mode	DCVS operating mode. Default is Force Voltage.

For a PSet example using VBT, see the DCVS PSet example function below.

For more information on PSet syntax, see TheHdw.DCVS.Pins.PSets.

DCVS PSet Example Function

This example function, AddPSet, creates a DCVS parameter set. To use the AddPSet function in Vbump testing, see Vbump Testing.

```
Function AddPSet(PinName As String, _
           PSetName As String, _
           BandwidthSetting As Double, _
           CurrentRange As Double, _
           SampleRate As Double, _
           SampleSize As Long, _
           SourceFoldLimit As Double, _
           MeterMode As tlDCVSMeterMode, _
           DCVSMode As tlDCVSMode, _
           MainVoltage As Double, _
           AltVoltage As Double _
           ) As Long
  On Error GoTo ErrorHandler
  Call TheHdw.DCVS.Pins(PinName).PSets.Add(PSetName)
  With TheHdw.DCVS.Pins(PinName).PSets.Item(PSetName)
    .BandwidthSetting.Value = BandwidthSetting
    .CurrentRange.Value = CurrentRange
    .Capture.SampleRate.Value = SampleRate
    .Capture.SampleSize.Value = SampleSize
```

```
        .CurrentLimit.Source.FoldLimit.Level.Value = SourceFoldLimit
        .Meter.Mode = MeterMode
        .Mode = DCVSMode
        .Voltage.Main.Value = MainVoltage
        .Voltage.Alt.Value = AltVoltage
    End With
Exit Sub
ErrorHandler:
    Dim Reply As Integer
    Reply = MsgBox("Error detected during AddPSet(). Stop to Debug?", vbYesNo, "UVS256 PSet
Checkout")
    If Reply = vbYes Then
        On Error GoTo ErrorHandler
        Stop
    Else
        On Error GoTo 0
    End If
    Resume Next
End Function
```

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UltraVS256 DCVS Programming : Using UVS256 in Concurrent Testing

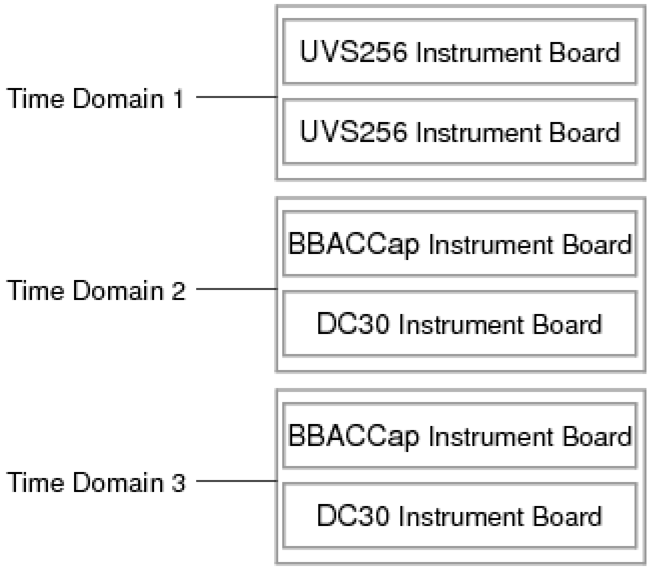
Using UVS256 in Concurrent Testing

To support concurrent testing, you can assign UVS256 channels to a time domain. This means that UVS256 instruments can be part of a concurrent pattern burst. However, only one pattern in the concurrent burst can contain UVS256 instruments in it at one time. This limitation means that UVS256 channels cannot be in more that one time domain for each concurrent burst.

Concurrent features for UVS256 include:

- A UVS256 board can be configured for one time domain
- Multiple UVS256 instruments can be grouped into a common time domain
- UVS256 is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with two UVS256 instruments assigned to its one time domain and other UltraFLEX instruments assigned to other time domains.



UVS256 Concurrent Testing Configuration Example

For more information, see About Concurrent Test.

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UltraVS256 Theory of Operation

UltraVS256 Theory of Operation

This section provides a theory of operation that includes a UVS256 hardware summary, test head connection information, a functional description with block diagrams, and output power specifications.

The following topics are available:

For more information on the UVS256, see:

- About the UltraVS256
- UltraVS256 DCVS Programming
- DCVS Debug Display
- UltraVS256 Examples

For information on UVS256 instrument performance specifications and standards, see UltraVS256 Specifications.

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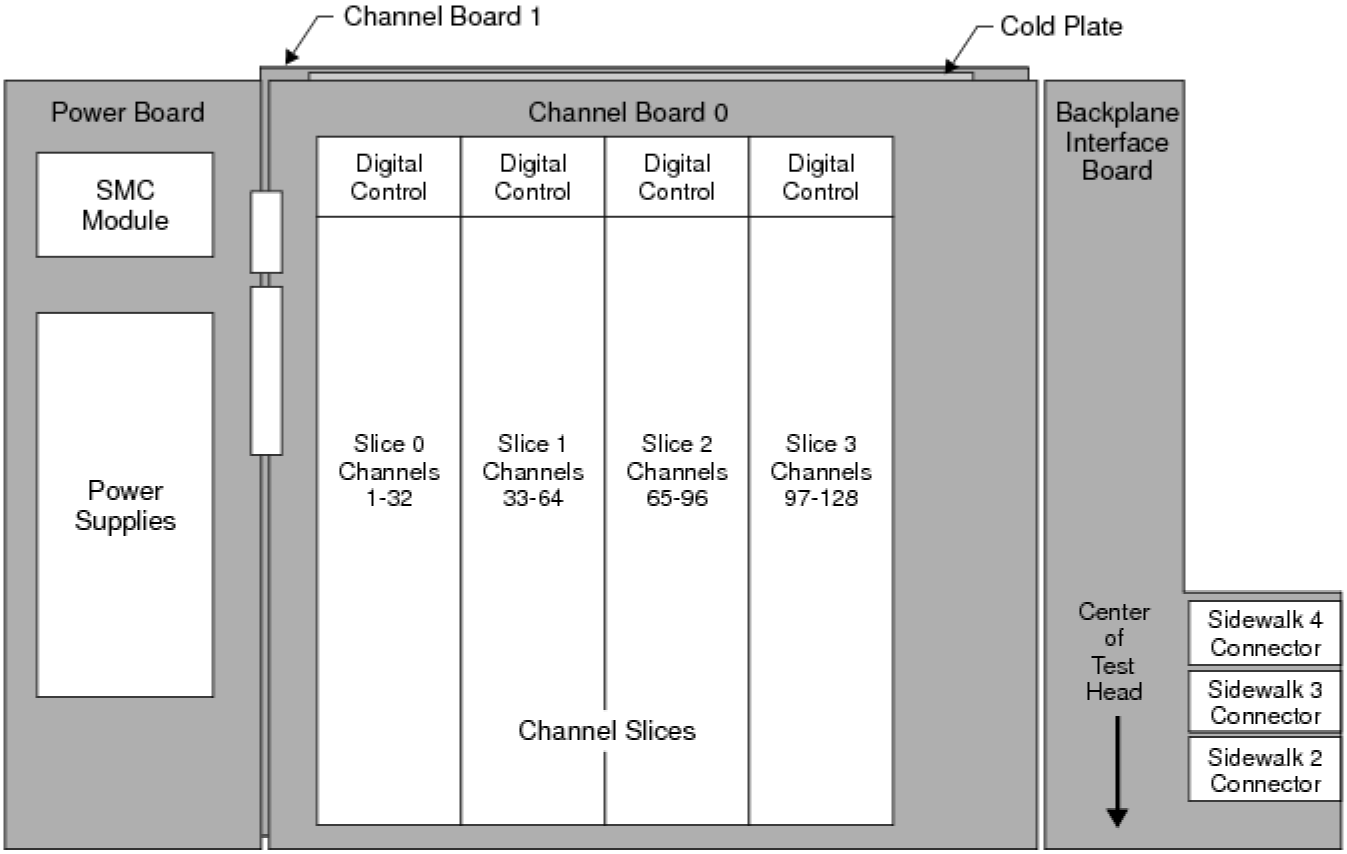
UltraVS256 Theory of Operation : UVS256 Board Hardware

UVS256 Board Hardware

The UVS256 instrument board assembly occupies a single slot in the UltraFLEX test head and consists of:

- Two identical 128 channel boards mounted on opposite sides of a liquid-cooled cold plate
- Four channel slices on each channel board that contain 128 channel DPS devices and the digital controls for these devices
- A backplane interface board (BIB) that provides the connections to the UltraFLEX backplane and manages signal flow between the test system and instrument boards
- A power board that converts the +48V system power to the various on-board power voltages that the UVS256 board requires

The following figure shows the layout of an UVS256 instrument board.



UVS256 Instrument Board Hardware Block Diagram

Channel Boards

The channel boards contain the primary DCVS instrument control and tester communication components as well as the output stage of the DCVS channels. The channel boards also have the following features:

- An interface to communicate with the databus of the tester computer
- A TCIO databus connecting to the instrument hardware
- A MoveLink bus for fast transfer of capture data to the Digital Signal Processing (DSP) hardware

Channel Slices

The UVS256 has eight channel slices, four on each channel board, that contain the analog circuitry of 256 DCVS power supplies. Each slice supports eight groups of four DCVS power supplies (channels). These supplies can be merged to increase the maximum output current and power delivered to the DUT. Eight slice digital control FPGAs, four on each channel board, support the module board hardware. Each slice FPGA supports eight groups of four channels. The DCVS power supplies' output stage is also on the main board, attached to the cooling assembly.

Each channel slice of 32 channels also includes the following resources:

- DDR2 memory
- Precision reference and buffering
- Current limiting and bandwidth control hardware
- ADC to meter voltage and current
- On-board calibration and checkers resources
- Parameter set (PSet) memory

DPS Channels

The UVS256 has 256 DCVS channels arranged in groups of four. Each channel has a single shunt ammeter circuit for both control loop functions and (user) current metering. The channel's meter can provide voltage or current measurements.

UVS256 also supports the Signals software model, which involves the TRIG microcode, and allows a record of equally timed samples to be captured as well as single measurements. Signals measurements for each slice of channels are stored in the DDR2 memory.

Each channel contains:

- A measure ADC
- An output connect for HF and HS, and a cal connect for HF
- Control loop
- Sense amplifiers
- Power output stages

Each group of four channels shares a DGS (low sense) input from the DUT.

Each channel group also include the following real-time controls:

•	Clamp Alarm

•	Open Kelvin Alarm

•	Over-Temperature shutdown

The clamp and Kelvin alarms from each DPS are sent to the digital control FPGA independently. In UVS256, groups of eight channels start at 1 and do not overlap. Each group of eight channels has a merge bus in common. DCVSMerged8 is the maximum merge configuration, where eight channels can deliver up to 5600mA to a DUT pin.

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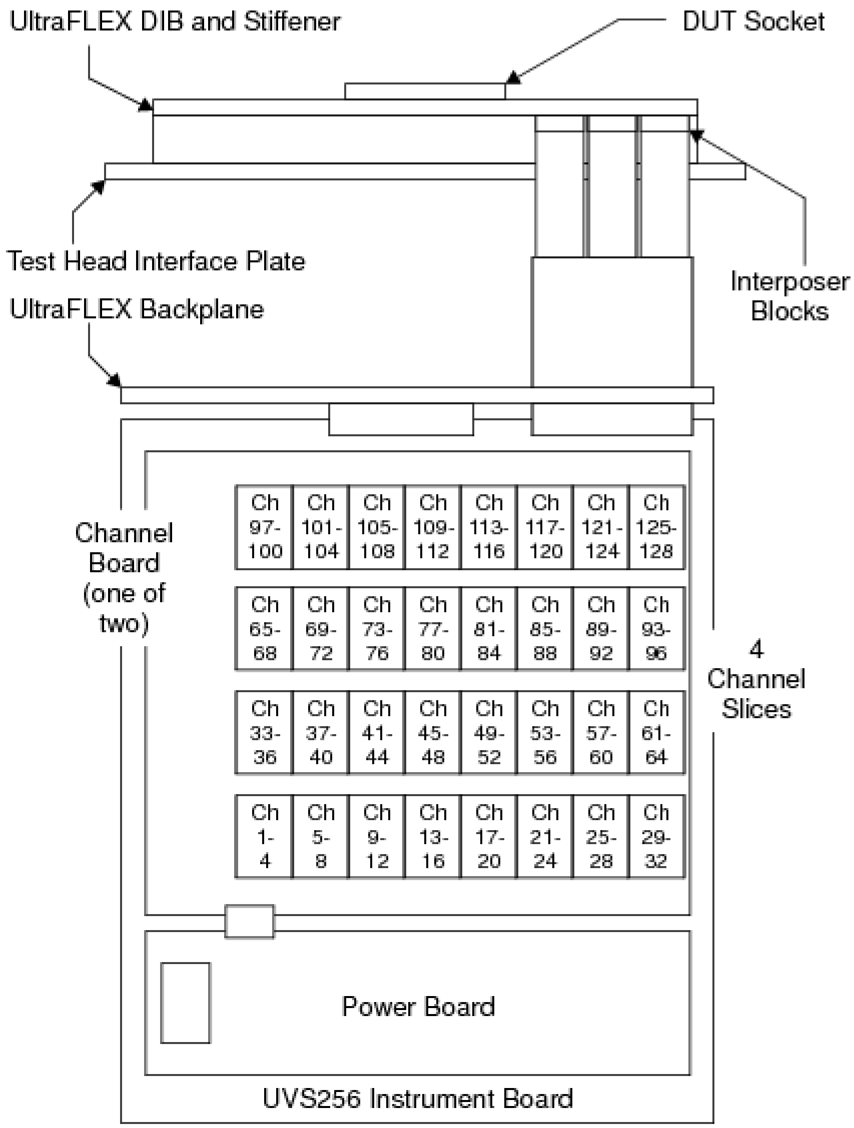
UltraVS256 Theory of Operation : Test Head Connections

Test Head Connections

This section shows where the boards are connected to the test head backplane.

The following figure shows how the instrument board is connected through interposer blocks to deliver power to the DUT.

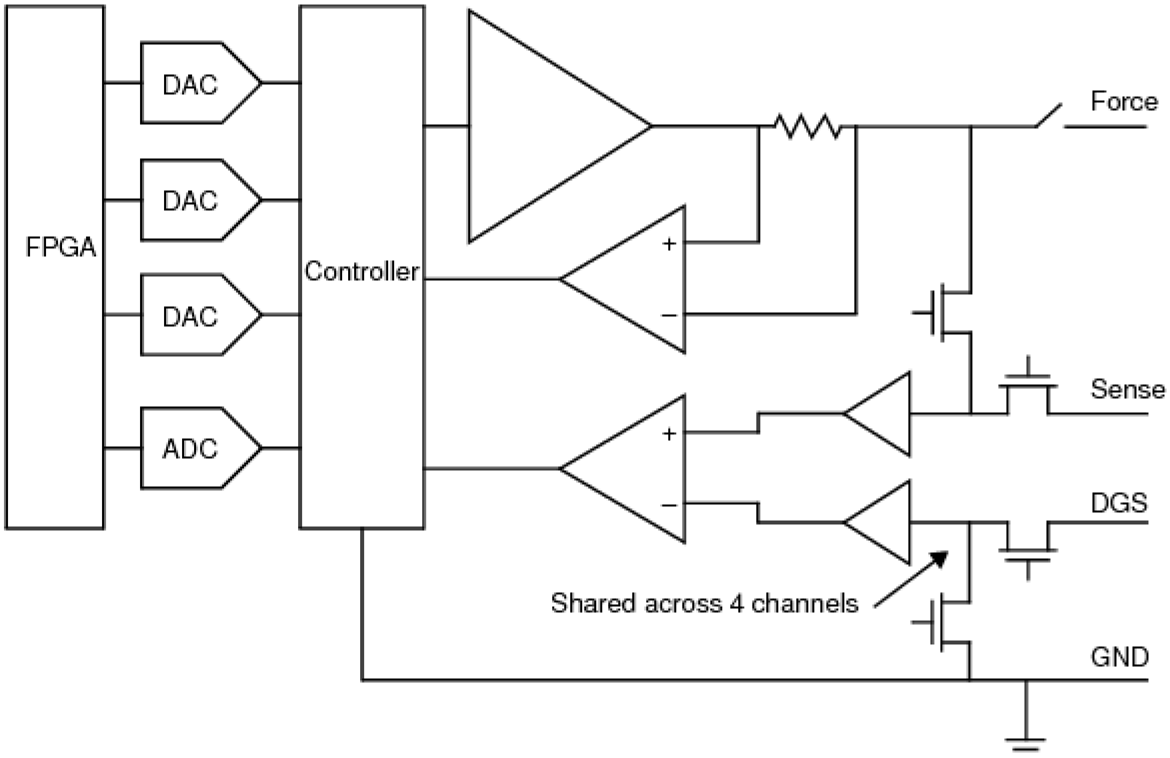


UVS256 Instrument Board Connection to the DUT

Several UVS256 boards can be installed in an UltraFLEX test head. A UVS256 can be installed in any available slot in the test head. Channels are connected to the DIB using three interposer blocks occupying sidewalks 2, 3, and 4 of the loadboard end.

The UVS256 test head interface features a high-performance connection to support high current to the DUT through a low impedance path. This interface allows each DCVS to respond rapidly to current transients, which helps reduce the need for large amounts of power supply bypass capacitance on the DIB. This saves space on the DIB for additional test sites, further enabling testing of multiple devices at once. For more information, see UltraVS256 DIB Design Guidelines.

The following figure shows a connection model for the UVS256 signals to the DUT.



UVS256 Channel Connections to the DUT

Each UVS256 channel has a DAC to provide analog voltage with two voltage value registers, main and alternate. It also has an ADC that is used for:

- Force voltage measurement and connected to internal GND or external DGS
- Current measurement

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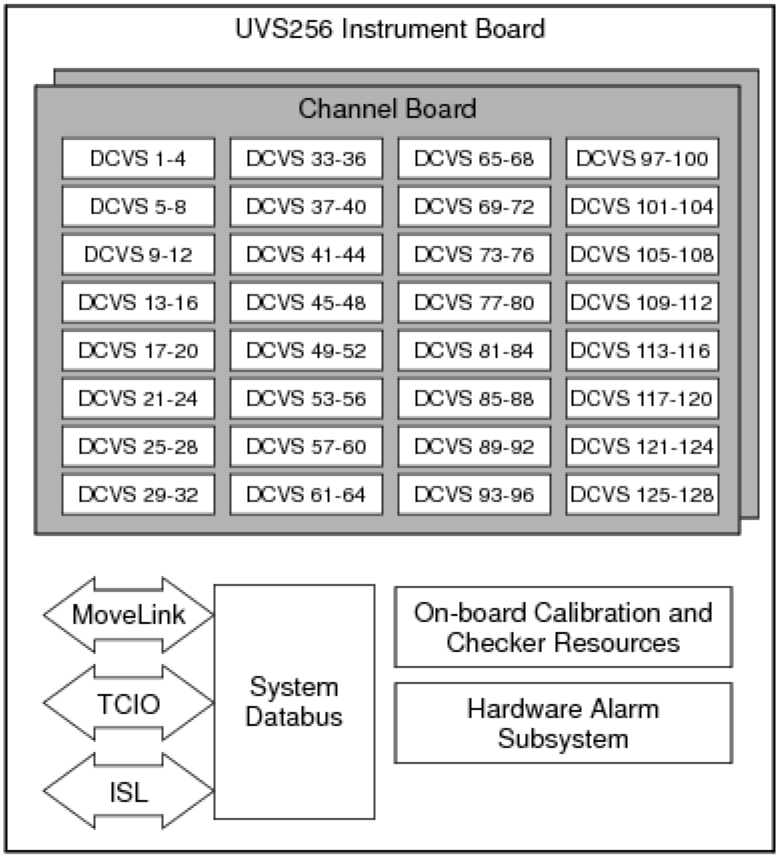
UltraVS256 Theory of Operation : Functional Description

Functional Description

This section describes how signals move through the UVS256 instrument board to the DUT from a programmer's point of view.

The UVS256 board consists of several functional subsystems:

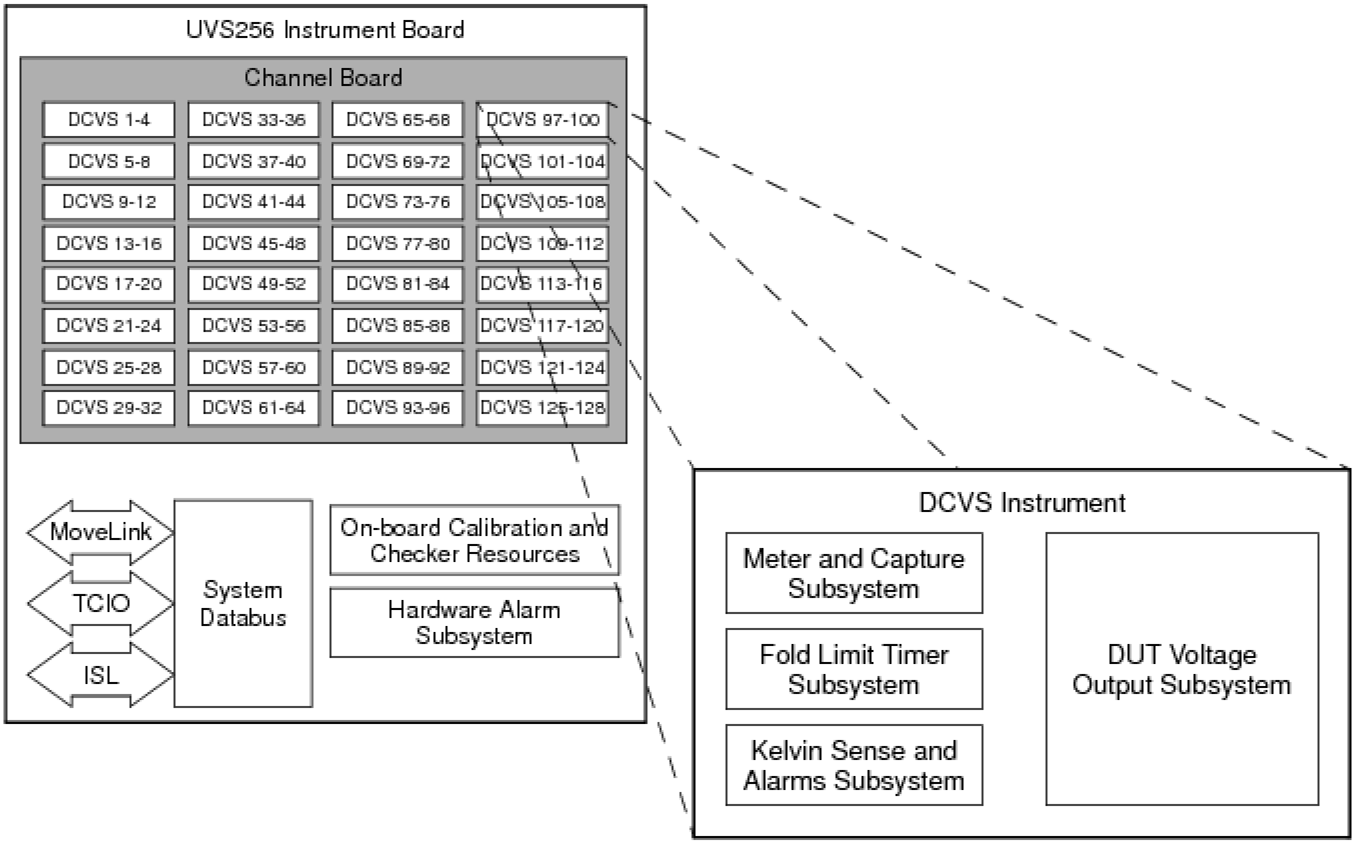
- 256 DCVS instruments, where two, four, or eight instruments can be merged to create fewer, more powerful instruments (up to 5.6A)
- Hardware alarm detection and response subsystem
- Tester databus communication (TCIO) subsystem
- MoveLink databus subsystem for fast transfer of capture data to the Digital Signal Processing (DSP) hardware
- ISL (Instrument Sync Link)
- Calibration and checkers subsystem



UVS256 Board Functional Overview (one of two boards shown)

Each DCVS instrument consists the following functional subsystems (see the following figures):

•	DUT Voltage Output Subsystem
•	Fold Limit Timer Subsystem
•	Metering and Capture Subsystem
•	Alarms Subsystem



DCVS Instrument Logical/Functional Overview

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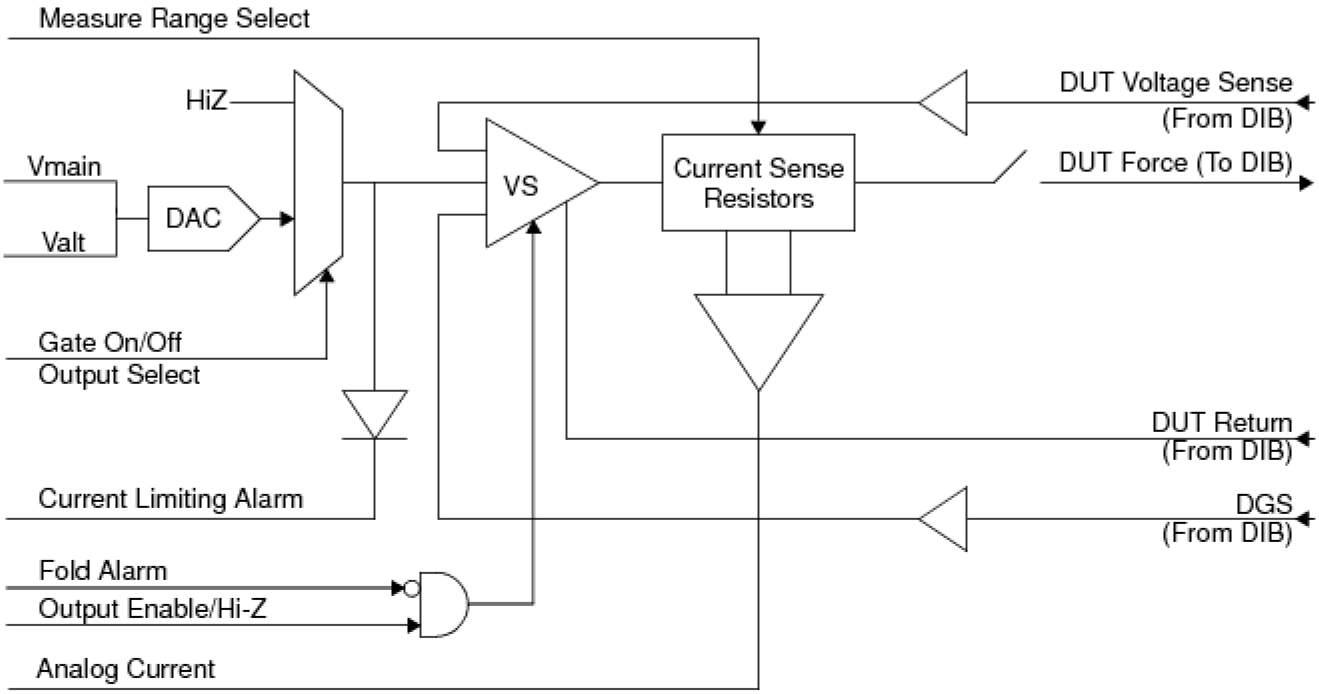
UltraVS256 Theory of Operation : Functional Description : DUT Voltage Output Subsystem

DUT Voltage Output Subsystem

The following figure shows a block diagram of the UVS256 voltage output subsystem.



UVS256 Instrument DUT Voltage Output Subsystem

DUT Voltage Sense Signal

The DUT voltage sense signal monitors the voltage at the DUT. When the voltage at the DUT exceeds the selected programmed value, the output driver drops its voltage, thereby decreasing the current sourced to the DUT. Similarly, when the sensed voltage at the DUT drops below the programmed value, the output driver increases its output voltage and sources more current to compensate for the drop at the DUT pins. This is an automatic and continuous process that uses the DUT sense feedback to control the power delivered to the DUT.

When the DCVS output current changes rapidly from a low level to a high one, you expect a voltage drop. The DCVS senses this drop and compensates for it. The DCVS output system bandwidth controls how quickly the DCVS output responds to changes in the DUT capacitor voltage.

The DCVS also features programmable output bandwidth settings. Bandwidth setting tells the analog hardware the characteristics of the load capacitance. By changing the bandwidth setting, you can program the DCVS output to respond faster or more slowly to changes in output current demand.

Programmable Current Fold Limit Signal

The current limiting signal and fold alarm help to control the maximum current that can be supplied to the DUT. For more information, see Fold Limit Timer Subsystem.

The DUT force and return lines make up the complete current loop bringing power to the device. They support the full current capability of the force lines. The return sense lines sense the power supply return path voltage at the DUT. So they are also part of the DUT voltage control loop that monitors the voltage across the DUT power pins.

Gating off the DCVS output sets the voltage output value to the voltage value sensed by the DGS DIB input signal. Ideally, this is the DUT ground voltage of 0.0V.

Embedded with the voltage output circuitry is the analog current and voltage meter circuitry. This includes the buffered DUT voltage sense input, current measure range select signal, analog current measurement hardware, and internal current signal shown in the figure above. For information on the functionality of these components, see Metering and Capture Subsystem.

Thermal Control Shutdown

The UVS256 output stage power amplifier has a thermal control shutdown feature to protect power supplies from damage. The thermal protection circuitry disables the output when the junction temperature reaches approximately 130C allowing the device to cool.

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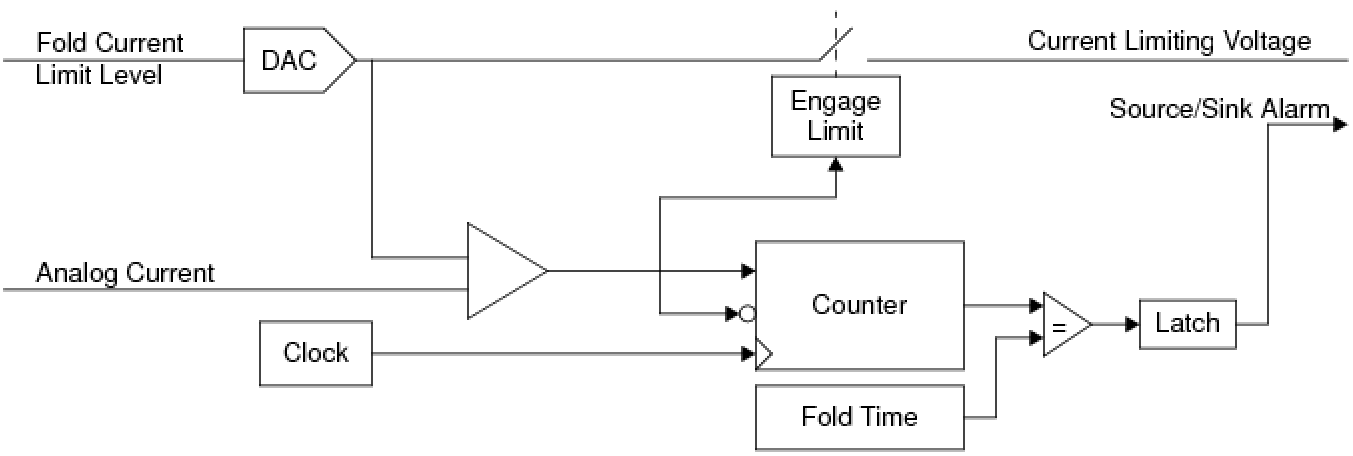
UltraVS256 Theory of Operation : Functional Description : Fold Limit Timer Subsystem

Fold Limit Timer Subsystem

Fold limit timer programming consists of setting the values of a DAC and counter. The following figure illustrates the concept behind a single UVS256 instrument's source and sink current limiting hardware.



Note: Limit Level Delta is specified in the UVS256 instrument board specification document.

UVS256 Instrument Source/Sink Fold Limiting Subsystem

You control current limiting functionality with VBT syntax. For sink/source current limiting, you can program a single fold current limit level and timer. As a part of the test setup, you can also program the current limit mode, which defines the fold limit timeout behavior.

While the instrument is gated on, the analog current output signal is compared against the fold limit level. If the analog current exceeds the programmed fold limit level, the current limiting voltage is engaged and a current limit timer is started. The output current is then limited to the fold limit level. If the timer value exceeds the fold limit time, a fold signal is set. Depending on the programmed current limit behavior mode, either the instrument will gate off, or it retains the programmed value when the fold limit timeout is exceeded.

Current fold limit timeout programmability is 0.3ms to 2000ms with a resolution of 0.1ms.

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UltraVS256 Theory of Operation : Functional Description : Metering and Capture Subsystem

Metering and Capture Subsystem

The UVS256 metering and capture subsystem consists of the following for each UVS256 channel:

- An ADC, which allows you to perform voltage or current metering
- A 1K strobe memory, which allows you to strobe voltage or current

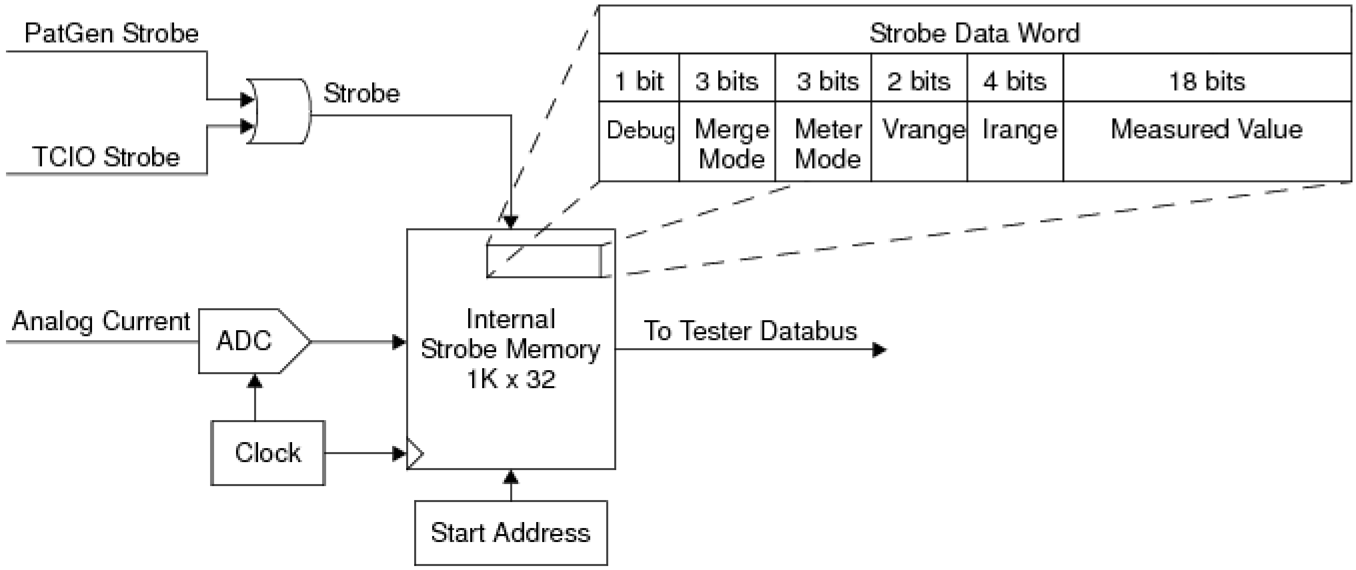
The strobe memory is a wrap-around memory implemented separately on each DCVS instrument channel. You can store up to 1024 samples in this memory until it is overwritten.

If you pass in a number that is greater than 1024 to the Meter.Strobe method's SampleSize parameter, IG-XL returns a run-time error because the maximum allowable number of samples is 1024.

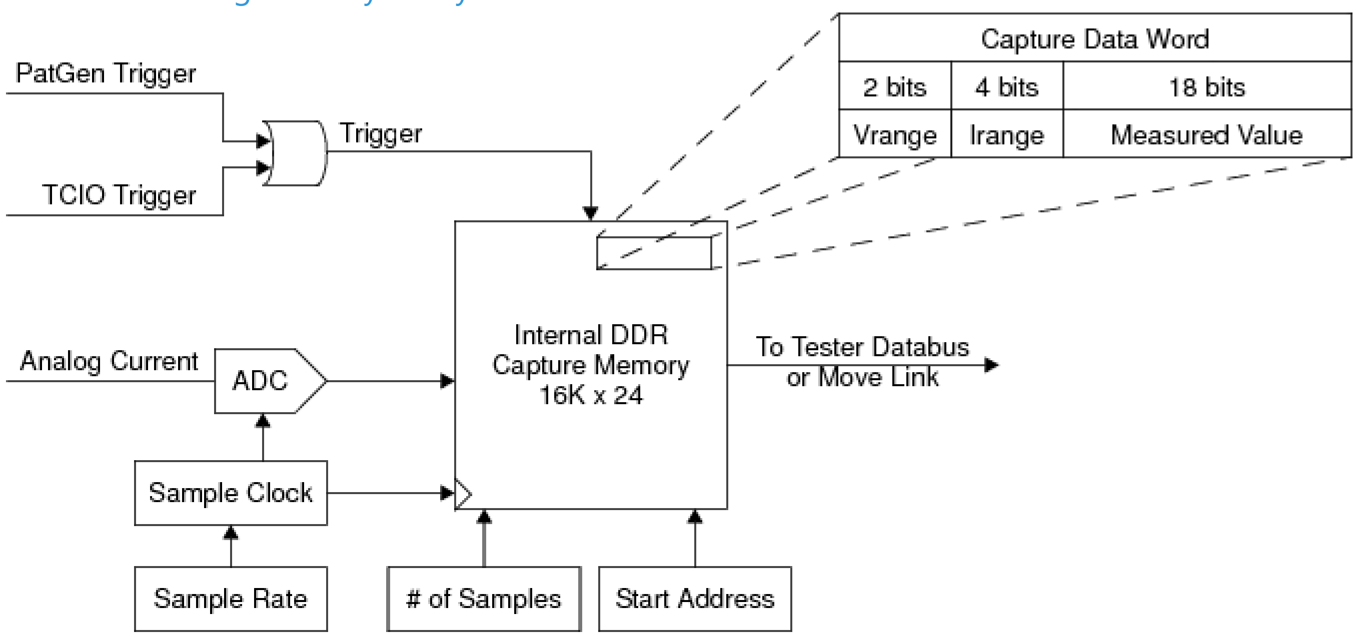
If you strobe more than 1024 samples in a pattern or exceed 1024 samples in multiple .Meter.Strobe statements, IG-XL returns a Memory Full alarm.

- A 16K capture memory for current profiling and debugging

You can use the UVS256 meter's ADC to convert either analog current or buffered DUT voltage sense signals to digital values. The digital values can then be stored in the capture memory. From there, the values can be read back to the host computer through the TCIO databus using the meter and capture programming interfaces. These features are shown in the following two figures.



UVS256 Metering Memory Subsystem



UVS256 Capture Memory Subsystem

A meter strobe is an averaged measurement of multiple ADC samples if the Filter is enabled and the Filter is set to less than 200K. A meter strobe is an ADC sample if the Filter is disabled or Filter is enabled and the Filter is set to 200K.

To make a meter measurement, you must set the meter mode, either voltage or current. The default meter mode is to measure current. Set the desired meter range before taking the measurement.

For a single-strobe measurement this is all the setup that is required before reading. To take multiple samples or to capture a waveform, you must program the number of samples. When capturing a waveform, you also program a sample rate and a signal name.

If a capture or meter voltage or current sample exceeds the presently selected meter range, IG-XL returns an overrange alarm.

The ADC has a fixed sample rate of 200kHz, and it can sample every 5s.

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UltraVS256 Theory of Operation : Functional Description : Alarms Subsystem

Alarms Subsystem

The UVS256 alarm subsystem hardware supports a DUT Kelvin alarm. For a list of alarm types and information on programming alarms, see Setting Alarms.

DUT Kelvin Alarm

Each UVS256 channel requires DUT sense Kelvin connections. The DUT Kelvin alarm monitors the voltage drop between the DIB sense points and the DUT sense points. It indicates a condition where the UVS256 control loop cannot be satisfied because the force-to-sense connection does not have a low enough impedance. The instrument triggers a DUT Kelvin alarm if the voltage difference between the VS output point (A) and the corresponding DUT voltage sense point (B) is greater than the preset Kelvin threshold value.

The DUT Kelvin threshold is 1.2V (hi-side). The DIB Kelvin threshold is 1.0V (lo-side).

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UltraVS256 Theory of Operation : Output Power

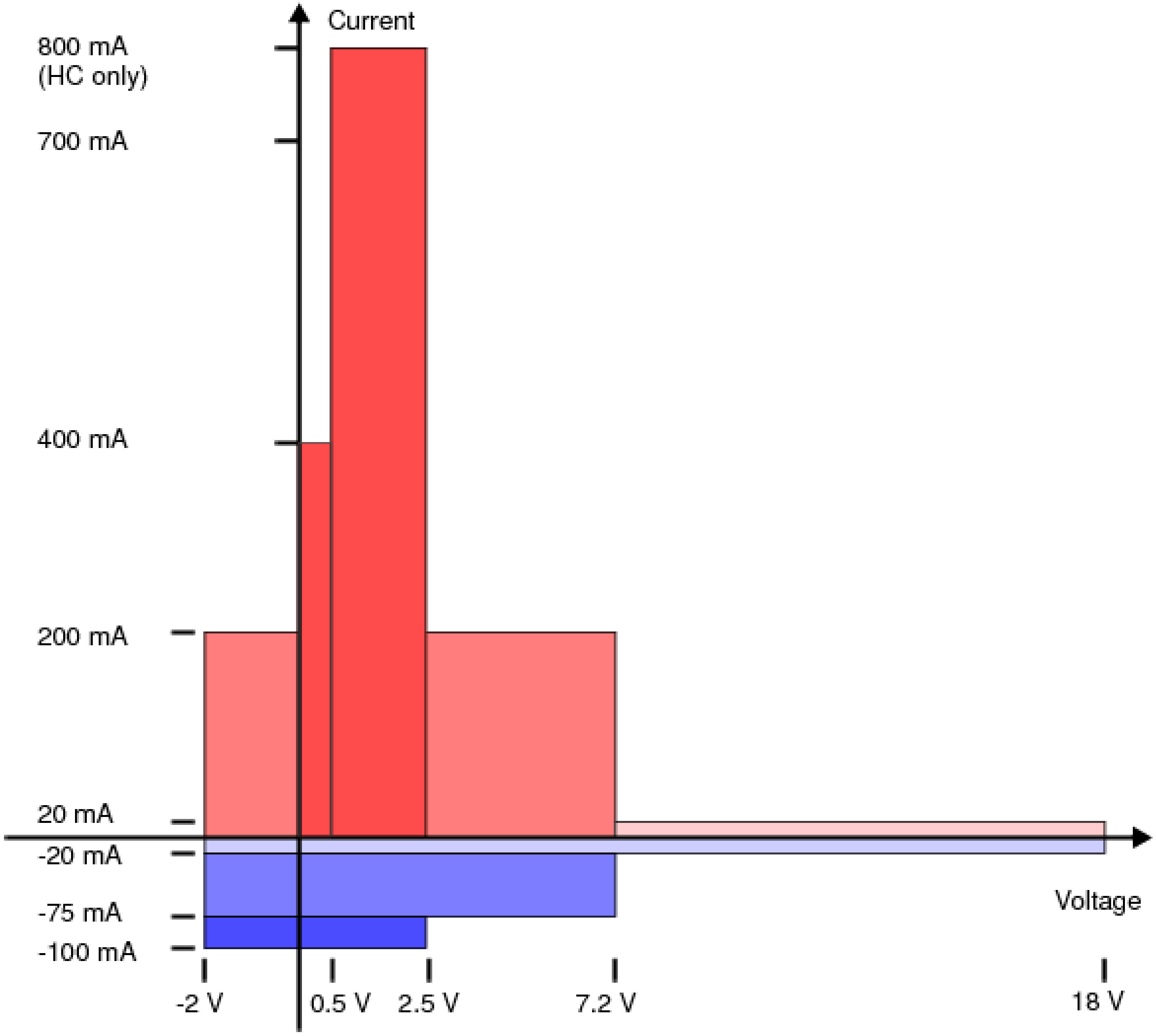
Output Power

This section discusses the output power capabilities and restrictions of the UVS256 board's instruments. It includes the following topics:

- UVS256 Operating Area
- Current Ranging
- Merging Channels

UVS256 Operating Area

The following figure shows the UVS256 operating area.



Voltage and Current Operating Area, Single Unmerged Channel

Each instrument on the UVS256 board can operate in four quadrants. You can:

- Source or sink to 20mA at voltages from -2V to 18V
- Source to 200mA at voltages from -2V to 7.2V
- Sink to -75mA at voltages from -2V to 7.2V
- Source current ranges greater than 200 mA at voltages from -2V to 2.5V (HC channels only)
- Sink current ranges to -100mA at voltages from -2V to 2.5V

UVS256 channels can operate from -2V to +18V. Maximum positive voltage is restricted based on the current range selected. On the highest current range, an unmerged channel can deliver 800mA with voltage limits of -2V to +2.5V.

The UVS256 instruments are not floating voltage supplies. As a result, their outputs cannot be stacked to provide greater output voltages. Moreover, you must wire the device ground sense (DGS) for each channel to ground. For more information, see UltraVS256 DIB Design Guidelines.

Current Ranging

UVS256 instruments have one set of current ranges. Each range determines both the largest current that can be sourced and the largest current that can be metered. Specified full-scale ranges per channel are from 800mA down to 4A, and additional ranges are available in merged modes. Channels using higher ranges (700mA and above) must be designated as high current (HC).

All specified current ranges have a limited tolerance for overrange that may allow metering beyond a given range. Metering more than 3% above the current range, however, will produce an overrange alarm.

UVS256 has a number of restrictions on current ranges. They include the following:

- Each group of four channels is limited to 2800mA output.
- The number of high current channels for each UVS256 is limited to 128 (64 channels per channel board)
- Standard current (SC) channels cannot access ranges above 200mA for a single channel or above 400mA for two merged channels

For more information, see the following:

- UVS256 High Current (HC) Channels
- Signal Delivery Path Current Limitation
- Channel Assignment Guidelines

Merging Channels

The UVS256 allows you to merge its DCVS outputs to provide up to 5.6A of current to a device pin. Although multiple UVS256 boards can be installed in a test system, board-to-board merging is not supported.

UVS256 channels can be merged when two conditions are met:

- You declare the channels in the channel map as a merged pin type.

- You have connected the respective output pins (HF, HS, LS, LF) for all channels in the merge group. For more information, see UltraVS256 DIB Design Guidelines.

Caution:
You must confirm that the DCVS merge configuration specified in the channel map matches the actual merge configuration of the DIB. If the merge configurations do not match, you can damage the DIB and the tester hardware during test program run time.
For more information, see UVS256 Merging.

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